



Land use land cover transformation and seasonal variations in land surface temperature with environmental implications in Gazipur Sadar Upazila, Bangladesh

Md. Rased Hasan Sojib^{a,*}, Md. Monabbir Hossain^a, Md. Al-Amin Khan^a,
Md. Abdul Baten^a, Md. Badiuzzaman Khan^a, F.K. Sayema Tanzia^b,
Mahmudul Hasan^a

^a Department of Environmental Science, Bangladesh Agricultural University, Mymensingh 2202, Bangladesh

^b Department of Environmental Science and Engineering, Jatiya Kabi Kazi Nazrul Islam University, Mymensingh 2220, Bangladesh

ARTICLE INFO

Keywords:

Urbanization
Land use/land cover change
Land surface temperature (LST) dynamics
Urban heat island (UHI) effect
Spectral indices

ABSTRACT

Rapid urbanization in Gazipur Sadar Upazila, Bangladesh, has significantly transformed land use land cover (LULC), contributing to rising land surface temperature (LST), and the intensification of urban heat island (UHI) effects. This study investigated the spatiotemporal changes in land use and land cover (LULC) and land surface temperature (LST) from 1999 to 2024. The changes in LST through different LULC classes were examined, and the correlation between various environmental indices and LST were assessed. This study also explored the spatial patterns and intensities of the urban heat island (UHI) effect and the urban thermal field variance index (UTFVI). Cloud-free (<5 %) Landsat 5 TM and Landsat 9 OLI/TIRS images from 1999, 2011, and 2024 (winter and summer seasons) were acquired from the USGS Earth Explorer and used to classify LULC (urban area, vegetation, waterbody, and bare land) and derive the LST. Supervised classification revealed that urban areas expanded by 93 % (from 61.3 km² to 118.3 km²), reducing vegetation by 9 % and water bodies by 45 %. Urban growth at an average rate of 2.7 % from 1999 to 2024, emphasized accelerated land conversion. Summer LST increased by 5.7 °C (from 26.8 °C in 1999 to 32.5 °C in 2024), with urban and bare land showing the highest temperatures. A strong positive correlation was observed between the LST and the normalized difference built-up index (NDBI), whereas with the normalized difference vegetation index (NDVI), the correlation was weakly negative. The urban thermal field variance index (UTFVI) revealed worsening ecological conditions in densely urbanized zones. The findings suggest the need for green infrastructure, water body conservation, and sustainable land-use policies to reduce heat stress and protect the environment.

1. Introduction

Rapid urbanization and socioeconomic development in various regions have led to substantial transformations in land use and land cover (LULC), significantly impacting local and regional climatic conditions (De Sherbinin et al., 2007; Cui and Shi, 2012). Urban

* Corresponding author at: Department of Environmental Science, Bangladesh Agricultural University, Mymensingh 2202, Bangladesh.
E-mail address: rasedese@gmail.com (Md.R.H. Sojib).

expansion often alters ecosystems, reduces biodiversity, and changes landscapes and the surrounding environment (Bahi et al., 2016; Liou et al., 2024). The United Nations (2018) projects that by 2050, 68 % of the global population will be vulnerable to surface urban heat island (SUHI) effects, driven by the rapid rise in LST (Fahmy et al., 2023). LST refers to the thermal radiation emitted from the Earth's surface as a result of absorbed solar energy, the intensity of which is often amplified in urban environments because construction materials with high thermal conductivity are used (Imran et al., 2021; Zhu et al., 2025). The LST plays a vital role across multiple scientific domains, including ecological monitoring, hydrological modeling, and climate change research, as it serves as a key indicator of surface-atmosphere energy exchange processes (Shahfahad et al., 2024). Urbanization brings significant development and prosperity to an area, but it also has negative effects in both the short and long term (Maimaitiyiming et al., 2014; Theodorou, 2022). One of the major long-term impacts is the rapid rise in LST, caused by changes in land use and urban growth (Mallick et al., 2008; Rasul et al., 2015).

The urban population is growing rapidly as more people move from rural areas to cities, looking for better job opportunities (Dutta et al., 2021). Such unplanned urban development alters land use patterns and disrupts surface heat balance, significantly increasing temperatures in urban areas (Sresto et al., 2022). In developing countries such as Bangladesh, the effects of such rapid urbanization are often more serious than in developed countries (Imran et al., 2021). High LST not only reduces people's comfort but also harms the environment and places pressure on both human health and ecosystems (Saha et al., 2024). As more people move from villages to cities in search of jobs and a better life (Dutta et al., 2021), the demand for urban services continues to rise adding to the pressure on land and worsening the heat problem (Fattah and Morshed, 2022).

Rapid urban population growth and expanding infrastructure are contributing to a significant decline in green and open spaces within cities (Zhang et al., 2024). LULC changes occur on the outskirts of cities because of the rapid pace of urban growth driven by population pressure and economic development (Bhat et al., 2017; Halder et al., 2023). LULC is a significant indicator of changes in land surface patterns, whereas land surface temperature (LST) is another important and widely used indicator for investigating environmental conditions and understanding climate change (Milesi et al., 2003; Grimm et al., 2008; Li et al., 2013; NourEldeen et al., 2020). It is important to understand the patterns of land use and land cover changes, as well as the thermal behavior of urban land surfaces, to manage the existing developed city areas, and it is equally important to plan the undeveloped parts of the city or cities in construction, with the aim of creating sustainable and eco-friendly urban environments (Zhang et al., 2013; Karakuş, 2019). Analyzing changes in urban morphology through LULC transformations and their influence on LST is essential for effective urban planning and enhancing thermal comfort (Cerqueira et al., 2024).

In recent years, many cities in Bangladesh have faced a sharp increase in LST, mainly due to unplanned urban growth (Hasnat, 2022). Cities such as Dhaka, Narayanganj, Chattogram, and Gazipur are expanding quickly with new buildings, roads, factories, and other infrastructure, but often without proper planning or enough green space (Ahmed et al., 2013; Faisal et al., 2022; Fattah and Morshed, 2022; Imran et al., 2021).

The distribution of land use and cover features, including buildings, roads, and green spaces in urban areas, is closely related to the LST, with each element contributing differently to heating and radiation (Bokaie et al., 2016). Changes in urban LULC accelerate climate change by reducing biodiversity and enhancing the urban heat island (UHI) effect (Faisal et al., 2022; Zhang et al., 2024). The urban heat island (UHI) effect refers to the overall warming of urban areas compared with their rural surroundings due to human activities. Satellite-derived LST are used to quantify the surface urban heat island (SUHI) effect (Zhou et al., 2014, 2018). High LST and the urban heat island (UHI) effect are among the most detrimental consequences of urbanization (Dou and Chen, 2017; Choudhury et al., 2019). The UHI effect results in elevated temperatures within urban areas, exacerbating environmental issues such as increased greenhouse gas emissions and air pollution, which in turn intensify LST and disrupt the spatial continuity of the surface thermal environment (Hu et al., 2024). UHI effect has also led to increased public health issues, such as respiratory problems (Halder et al., 2023), and presents a significant challenge for town planners (Saha et al., 2021). Quantitative studies on LST and UHIs across different land surfaces provide valuable insights for developing urban heat management plans (Liang et al., 2021; Zhang et al., 2024). Increased LST and the UHI effect are key challenges of urbanization, and quantitative studies on LST and UHIs can help city planners and urban researchers develop urban heat management plans (Choudhury et al., 2021; Dou and Chen, 2017; Liang et al., 2021). The impermeable surfaces of urban areas absorb solar heat during the day and release it at night, disrupting the surface energy balance and leading to an increase in the LST (Yadav et al., 2017; Wang et al., 2025). The reduction in vegetation in urban areas results in a decrease in air humidity, whereas impermeable surfaces capture excessive heat, collectively contributing to an increase in urban land surface temperature (Hass et al., 2016). The constant loss of green areas due to rapid urban expansion activates the urban heat island effect and affects urban microclimates because decreased forest cover and restricted open soil fail to absorb the extra heat created by impermeable surfaces (Jesdale et al., 2013; Jamali et al., 2022; Arshad et al., 2022). Changes in local temperatures negatively affect urban health, degrade air and water quality, disrupt precipitation patterns, disturb biological balance, and change landscapes (Hasnat, 2022).

Field surveys and manual data collection to monitor changes in LULC and LST over time can be time-consuming, laborious, and costly (Hasnat, 2022). Sometimes, this type of data can be biased owing to a small sample size, or it may be incomplete if certain areas are inaccessible and need to be skipped (Goldblatt et al., 2021). The gradual advancement of satellites and tools such as remote sensing has made it easier to collect data and analyze essential information in a relatively short time with high accuracy and at a lower cost (Hussain et al., 2022; Silva López et al., 2022). Many researchers in Bangladesh have investigated LULC changes and their effects on LST over time in places such as Dhaka city, Rajshahi city, Chittagong city, Cumilla city, Khulna city (Choudhury et al., 2019; Rakib et al., 2020; Faisal et al., 2021; Imran et al., 2021; Hasnat, 2022; Mia et al., 2022; Halder et al., 2023; Saha et al., 2024). Over the past few decades, this area has experienced significant urban growth and industrial development, with numerous factories and industries being established. The presence of nearly 1500 garment factories has spurred rapid growth and urbanization in Gazipur, transforming it into a key economic hub in Gazipur Sadar Upazila (Ahmed et al., 2018). Despite experiencing substantial urban expansion and the

establishment of numerous factories over the past few decades, the spatiotemporal dynamics of LULC and LST in this area remain largely unexamined. Our understanding of the impacts of unplanned industrialization and urbanization on the local thermal environment remains limited. However, this type of research is not widely available for Gazipur Sadar Upazila, the administrative region of the Gazipur district in Bangladesh.

The main objectives of this study were to: (i) measure the spatial-temporal changes in land use and land cover (LULC) and land surface temperature (LST) between 1999 and 2024, (ii) analyze LST variation across different LULC classes, (iii) examine the correlations among various ecological indicators and LST, and finally, (iv) explore the spatiotemporal patterns and intensity of urban heat island (UHI) and urban thermal field variance index (UTFVI).

1.1. Study area

Gazipur Sadar Upazila, located in the Gazipur District of Bangladesh, covers an area of approximately 457.67 km². Geographically, it spans between 23°53' and 24°11' North latitude and 90°20' and 90°35' East longitude (Fig. 1). It is bounded by Sreepur Upazila to the north; Savar Upazila, Uttara Thana, and Rupganj Upazila to the south; Kaliganj and Rupganj Upazilas to the east; and Kaliakair and Savar Upazilas to the west. The upazila was established in 1983 through the merger of Joydebpur and Tongi. Topographically, Gazipur Sadar lies within the Madhupur Tract, an elevated Pleistocene terrace region. It is characterized by red-brown loamy soils and gently undulating terrain. This elevation provides relative resistance to seasonal flooding, making it attractive for industrial expansion and urban development. The area experiences a subtropical monsoon climate, with a hot and humid summer, a monsoon season from June to September, and a mild winter. The Average annual precipitation ranges between 1800 mm and 2400 mm, and the mean monthly temperature varies from approximately 12 °C in January to 35 °C in April or May (BMD, 2023).

Gazipur Sadar has undergone significant urban and industrial expansion in recent decades. It serves as a major industrial hub, housing numerous garment factories and manufacturing units that contribute substantially to the national economy. According to the Bangladesh Bureau of Statistics (BBS), the population of Gazipur Sadar was 228,321 in 2011 (BBS, 2011) and increased to 345,801 in 2022 (BBS, 2022). The area features dense urban settlements in Tongi and Joydebpur, with peri-urban expansion spreading toward the outskirts. Many residents work in local industries, while many workers travel to Dhaka every day because housing in Gazipur is more affordable. The distance between the two cities is approximately 33 km.

Historically, the region was part of the forested Bhawal Pargana, known for its dense vegetation (Roy et al., 2021). However, rapid

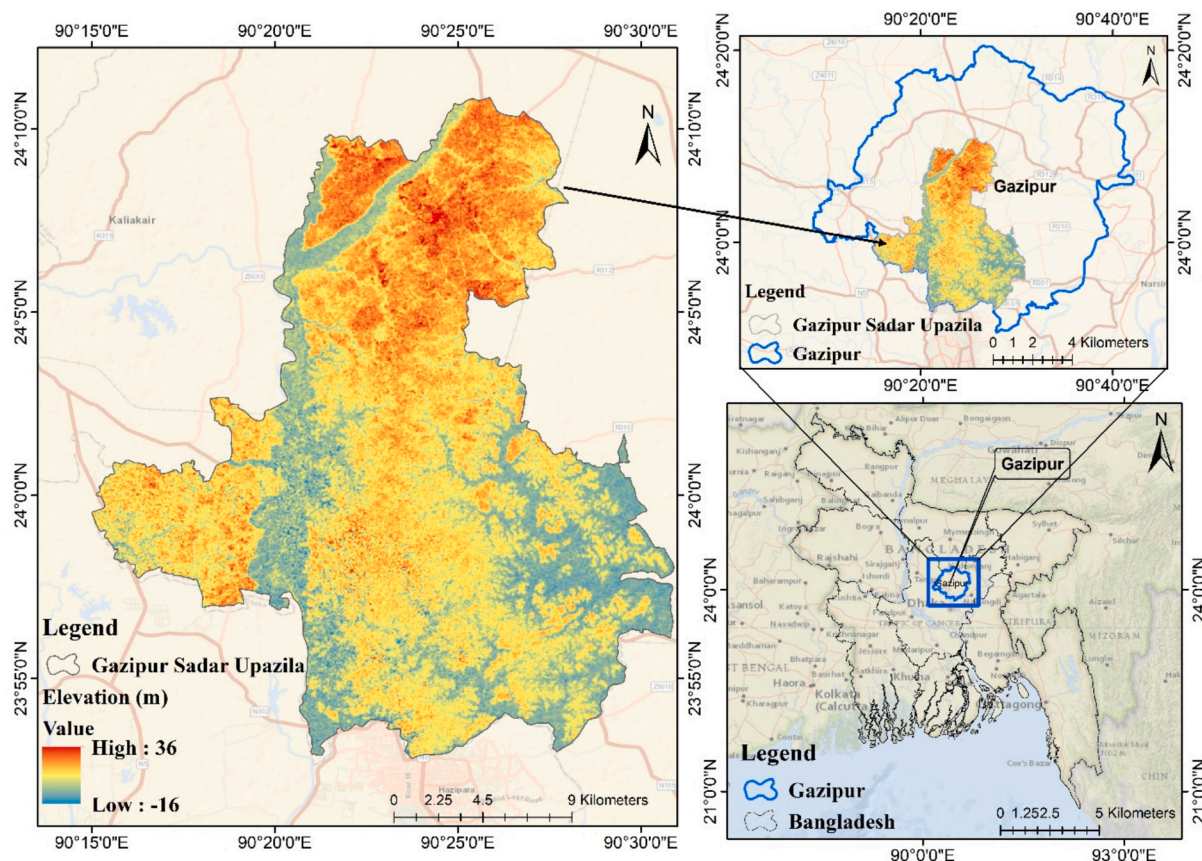


Fig. 1. Study area of Gazipur Sadar Upazila, Bangladesh.

urbanization has brought several environmental and social challenges. These include inadequate solid waste management, frequent waterlogging due to poor drainage infrastructure, the encroachment and degradation of water bodies, and a shortage of public recreational spaces. The upazila is also characterized by a growing population of low-income communities that often lack access to essential urban services, such as clean water, sanitation, healthcare, and education. This socioenvironmental context underscores the urgency of monitoring land use changes and associated thermal impacts in the region.

2. Materials and methods

2.1. Data acquisition

Multispectral Landsat data were acquired from the USGS (<https://earthexplorer.usgs.gov>) database to examine the changes in LULC classes and variations in LST. This study used two satellites, Landsat 5 TM and Landsat 9 OLI/TIRS, which were collected from Path 137/43 with a consistent spatial resolution of 30 m. Data were collected for both the winter and summer seasons. The maximum cloud coverage was set to less than 5 %, and all the selected images had minimal cloud presence, ensuring clear and reliable data over the study area (Table 1).

3. Methods

3.1. Research design flowchart

The research framework is illustrated in a flowchart that outlines the overall process (Fig. 2).

3.2. Image processing

ArcGIS (v10.8) software was used to process and analyze the satellite imagery. Image processing included atmospheric, radiometric, and geometric corrections. Atmospheric correction is a crucial step that needs to be monitored because the presence of gases, solids, and liquid particles in the atmosphere can affect the signal measured by satellites. Radiometric corrections were applied to correct the data for sensor irregularities, unwanted sensor or atmospheric noise, and clouds (Faruq et al., 2016). Image processing was carried out to eliminate systematic errors from the images, ensuring a smoother and more accurate classification process (Hasnat, 2022).

3.3. Classification of land use and land cover

The supervised maximum likelihood classification (MLC) technique was applied to distinguish various LULC classes. Four primary LULC types, including urban areas, vegetation, water bodies, and bare land, were selected (Table 2). Various recent studies have identified these crucial characteristics for assessing LULC dynamics and their environmental impact (Rakib et al., 2020; Kafy et al., 2020; Das et al., 2021; Faisal et al., 2021). The supervised maximum likelihood classification approach depends on training sites chosen according to the user's field knowledge and skill (Mahata et al., 2024).

3.4. Accuracy analysis and LULC map validation

Accuracy assessment is a crucial step in evaluating the performance of a classification method, particularly in remote sensing and land use and land cover (LULC) mapping. It involves comparing the results of a classification with reference data to determine how well the classification matches the actual ground conditions. In this study, the reference classification provided by the Google Earth Map considers accurate and standard baselines. The time slider feature was used to compare historical Google Maps with the classified images from 1999 to 2024. The sample data from each classified image and its corresponding map were compiled into an error matrix for analysis (Chughtai et al., 2021; Maxwell et al., 2021).

The accuracy assessment calculates how accurate each class on the map is, along with the overall accuracy. To do this, 200 stratified random sample locations were chosen from the classified image and then compared with the relevant Google Earth map to generate error matrices and determine the accuracy for each class and overall.

Table 1
Summary of collected satellite data.

Satellite	Sensor	Path/Raw	Spatial Resolution (m)	Cloud Cover (%)	Date	Season	Purpose of Use
Landsat 5	TM	137/43	30	0	18th December 1999	Winter	LST
Landsat 5	TM	137/43	30	0	21th March 1999	Summer	LULC and LST
Landsat 5	TM	137/43	30	0	17th Jan 2011	Winter	LST
Landsat 5	TM	137/43	30	2	7th April 2011	Summer	LST and LULC
Landsat 9	OLI and TIRS	137/43	30	0.06	30th December 2024	Winter	LST and LULC
Landsat 9	OLI and TIRS	137/43	30	4.20	26th April 2024	Summer	LST

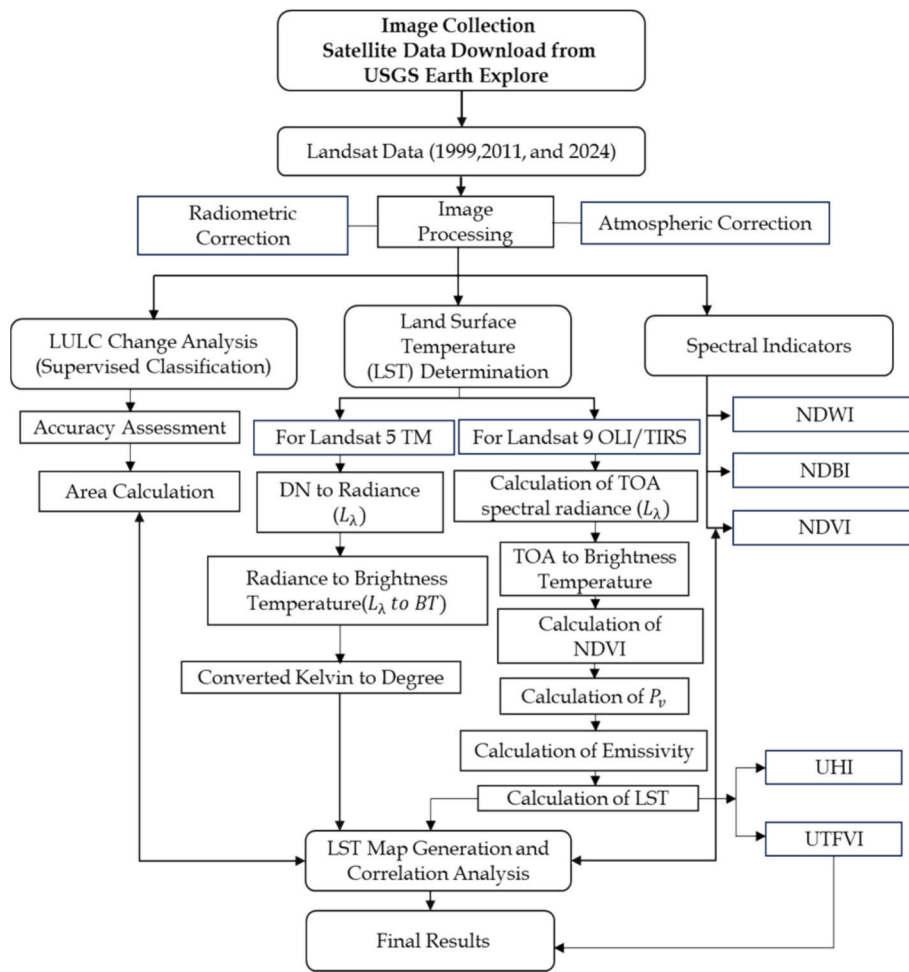


Fig. 2. Methodological flowchart of this study.

Table 2

Descriptions of LULC categories.

LULC Type	Description
Urban Area	All infrastructure-residential, commercial, mixed-use and industrial areas, villages, settlements, road networks, pavements, and man-made structure.
Vegetation	Deciduous forest, mixed forest lands, palms, conifer, scrub and others, Cultivable land, crop fields, fallow lands and vegetable fields.
Water Body	River, Lakes, ponds, canals, low-lying areas, marshy land, permanent/seasonal wetlands.
Bare Land	Bare soils, fallow land, construction sites, developed land, excavation sites, open space.

The user's accuracy and producer's accuracy measure how accurately a specific region has been classified. The omission error, associated with producer accuracy, represents the proportion of observed locations in the field that were not correctly identified in the reference map (Maxwell et al., 2021). Overall accuracy represents the proportion of pixels that were correctly mapped in comparison to the reference locations on the ground. Kappa coefficient values range from 0 to 1. A value greater than 0.80 (80 %) indicates strong agreement and good accuracy, values less than 0.40 and 0.80 (40–80 %) suggest moderate agreement, and values below 0.40 indicate poor agreement (Congalton, 2001; Lillesand et al., 2015). The user's accuracy, producer's accuracy, overall accuracy, and kappa coefficient ($\bar{K}$) percentage equations were computed via Eqs. (1)–(4) (Hasnat, 2022).

$$\text{User accuracy} = \frac{\text{Number of correctly classified pixels in each category}}{\text{Total number of classified pixels}} \times 100 \quad (1)$$

$$\text{Producer Accuracy} = \frac{\text{Sample Correctly identified in the column}}{(\text{The column total})} \times 100 \quad (2)$$

$$\text{Overall Accuracy} = \frac{\text{Total number of correctly classified pixels (Diagonal)}}{\text{Total number of reference pixel}} \times 100 \quad (3)$$

$$\text{Kappa Coefficient}(\hat{K}) = \frac{(\text{TS} \times \text{TCS}) - \sum (\text{Column Total} \times \text{Row Total})}{\text{TS}^2 - \sum (\text{Column Total} \times \text{Row Total})} \times 100 \quad (4)$$

where:

- TS = Total number of samples
- TCS = Total number of corrected samples

3.5. Land surface temperature calculation

Land surface temperature (LST) is influenced by the amount of heat radiated from the Earth's surface, which varies based on the basis of different types of land use and land cover (LULC) (Kayet et al., 2016). The thermal bands collected from various Landsat series images are used for calculating the LST (Mahata et al., 2024). The thermal data in Landsat sensors are represented in the form of digital numbers (DNs) (How Jin Aik et al., 2020). The LST calculations for this study were performed via Band 6 of the Landsat-5 TM and Band 10 of the Landsat-9 OLI. Those digital numbers (DNs) are converted to radiance values according to the formulae appropriate for the sensor that generated the data in the sensor manual. In the case of Landsat 8 and Landsat 9 OLI/TIRS, Band 10 is the favored option for the LST calculations over Band 11 as it has improved radiometric calibration and less atmospheric interference sensitivity.

3.5.1. Land surface temperature calculations for Landsat 5 TM, and Landsat 9 OLI/TIRS

Using ArcGIS 10.8, the thermal radiance was calculated manually from the digital values and then converted to temperature via the appropriate formula (Eqs. (5b-6)). To calculate the LST for Landsat 5 TM, and Landsat 9 images, the following steps are typically involved:

Step 1: Radiance calculation (in $\text{W}/\text{m}^2 \cdot \text{sr} \cdot \mu\text{m}$)

The digital numbers (DN) in Band 6 and Band 10 (thermal infrared band) are converted to radiance. The solar energy captured by the Landsat 9 OLI/TIRS and Landsat 5 (TM) sensors was manually converted into top of atmosphere (TOA) radiance via Eqs. (5a) for Landsat 9 OLI/TIRS, and (5b) for Landsat 4–5 TM (Hasnat, 2022).

$$L_{\lambda_i} = \text{ML} \times Q_{\text{cal}} + \text{AL} - O_i \quad (5a)$$

$$L_{\lambda_i} = \frac{Q_{\text{cal}} - Q_{\text{cal_min}}}{Q_{\text{cal_max}} - Q_{\text{cal_min}}} \times (L_{\lambda_i(\text{max})} - L_{\lambda_i(\text{min})}) + L_{\lambda_i(\text{max})} \quad (5b)$$

where,

- Q_{cal} = calibrated DN value,
- $Q_{\text{cal}(\text{max})}$ and $Q_{\text{cal}(\text{min})}$ = minimum and maximum DN values
- $L_{\lambda_i(\text{max})}$ and $L_{\lambda_i(\text{min})}$ = minimum and maximum radiance values (specific to Landsat 5 TM, and Landsat 9).

Step 2: Brightness temperature calculation (Kelvin) from the spectral radiance

To calculate the brightness temperature (T_b) from the radiance, Planck's Law (the inverse relationship between the radiance and temperature) is used. The band radiances calculated via Eqs. (5a) and (5b) were transformed into top-of-atmosphere (TOA) brightness temperatures (T_b) via the thermal constants provided in the image metadata. This conversion process follows a similar formula for both Landsat 9 OLI/TIRS and Landsat 5 (TM) (Ramaiah et al., 2020). The conversion was calculated via eq. (6), as shown below:

$$T_b = \frac{K_2}{\ln\left(\frac{K_1}{L_{\lambda_i}} + 1\right)} - 273.15 \quad (6)$$

where,

- T_b = Brightness temperature in Kelvin (K),
- $K_1 = 607.76$ (Landsat 5 TM) for Thermal Band 6 or $K_1 = 799.03$ (Landsat 9 OLI) for Constant Band 10
- $K_2 = 1260.56$ (Landsat 5 TM) for Band 6; or $K_2 = 1329.24$ (Landsat 9 OLI) for Constant Band 10
- L_{λ_i} = Radiance calculated in the previous step

Step 3: Normalized difference vegetation index (NDVI)

The NDVI was determined to assess the relationship between the top of atmosphere (ToA) reflectance values of the red and near-infrared (NIR) bands (Tucker, 1979). NDVI was calculated by applying Eq. (7) (Broge and Mortensen, 2002).

$$NDVI = \frac{NIR - RED}{NIR + RED} \quad (7)$$

where,

- NIR is the reflectance value in the near-infrared band (Band 4 for Landsat 5 TM and Band 5 for Landsat 9 OLI/TIRS).
- RED is the reflectance value in the red band (Band 3 for Landsat 5 TM and Band 4 for Landsat 9 OLI/TIRS).

Step 4: Calculating the proportion of vegetation

The proportion of vegetation is calculated using Eq. (8) (Rakib et al., 2020; Haldar et al., 2023). In this calculation, the lowest and highest NDVI values were taken into consideration:

$$P_v = \left(\frac{NDVI - NDVI_{min}}{NDVI_{max} - NDVI_{min}} \right)^2 \quad (8)$$

where:

P_v = Vegetation proportion, $NDVI$ = DN values from the NDVI image, $NDVI_{min}$ = Minimum DN values from the NDVI image, and $NDVI_{max}$ = Maximum DN values from the NDVI image.

Step 5: Land surface emissivity (LSE)

The land surface emissivity (LSE) is calculated via Eq. (9) (Haldar et al., 2023; Mahata et al., 2024).

$$\varepsilon = .004 \times P_v + 0.986 \quad (9)$$

where,

ε = Land surface emissivity, P_v = Vegetation proportion, 0.004 and 0.986 correspond to correction values of the equation.

Step 6: Land surface temperature (LST)

Land surface temperature (LST) is a critical index for evaluating climate change and global warming because of its significant influence on environmental systems. The LST is the radiative temperature, which is calculated from the top of atmosphere brightness temperature, the wavelength of emitted radiance, and the land surface emissivity (Abir et al., 2021). Following the preceding steps, the LST can be derived at this stage via eq. (10).

$$LST = \frac{T_b}{1 + \left(\frac{\lambda \times T_b}{\rho} \right) \times \ln(\varepsilon)} \quad (10)$$

Where,

- T_b = Brightness temperature ($^{\circ}C$)
- λ = Wavelength emitted radiance (the value of λ Landsat 9: for a band is 10.8 μm)
- ε = Land surface emissivity
- $\rho = \frac{h \times c}{k} = 1.4388 \times 10^{-2} mk = 14388k$, Planck's constant (h) = $6.626 \times 10^{-34} js$, Boltzmann constant (k) = $1.38 \times 10^{-23} JK$,
- velocity of light (c) = $(2.998 \times 10^8 ms^{-1})$

3.6. Environmental spectral indices

To better understand the relationship between LULC changes and LST, multiple spectral indices, including the NDVI, NDBI, and NDWI, were calculated via different bands from Landsat imagery (Mahata et al., 2024). Several prior studies, such as those by Rousta et al. (2018) and Santhosh and Shilpa (2023), utilized these indices to explore the connection between LULC dynamics and LST.

The NDVI technique, derived from multispectral remote sensing, is commonly used to distinguish various vegetation types, categorize land cover, and identify features such as water bodies and bare land (Haldar et al., 2023; Halder et al., 2022). The NDVI was applied to assess changes in both the quality and extent of vegetation over time and space. The index was calculated via Eq. (11) (Das et al., 2021):

$$NDVI = \frac{NIR - RED}{NIR + RED} \quad (11)$$

In this formula, NIR represents the near-infrared band (μm), and RED denotes the red band (μm). The NDVI values range from -1.00 to 1.00 , with values closer to -1.00 indicating water bodies and those approaching 1.00 indicating healthy vegetation and forest cover (Halder et al., 2021).

The spatial extent of built-up areas was quantified via the NDBI. The near-infrared (NIR) and shortwave infrared (SWIR) wavelengths are utilized in this formulation to ensure that water bodies have negative pixel values, whereas built-up areas are represented by positive values (Herrera et al., 2019). The index was derived via Eq. (12) (Verma and Garg, 2021):

$$NDBI = \frac{SWIR - NIR}{SWIR + NIR} \quad (12)$$

In Eq. (12), SWIR denotes the shortwave near-infrared band (μm). The NDBI values range between -1.00 and 1.00 , with negative values indicating water bodies and positive values signifying built-up areas (Mahata et al., 2024).

The normalized difference water index (NDWI) is recognized as an important index for mapping and identifying water bodies. The NDWI was utilized to detect surface water features, including ponds, lakes, rivers, and wetlands. The calculation of the NDWI was based Eq. (13) (Ashwini and Sil, 2022):

$$NDWI = \frac{NIR - SWIR}{NIR + SWIR} \quad (13)$$

NDWI values range from -1.00 to 1.00 , with values greater than 0.50 typically indicating the presence of a water body.

3.7. Urban heat island (UHI) and urban thermal field variance index (UTFVI) calculations

For UHI analysis, UHI maps were created via Eq. (14) (Halder et al., 2022). Both UHI and UTFVI maps were generated using the “Raster Calculator” tool in ArcGIS (v10.8).

$$UHI = \frac{LST - LST_{\text{mean}}}{SD} \quad (14)$$

where, LST_{mean} is the average land surface temperature ($^{\circ}\text{C}$) and SD represents the standard deviation of the land surface temperature ($^{\circ}\text{C}$).

The urban heat island (UHI) effect has been widely studied via the urban thermal field variance index (UTFVI) (Zhang et al., 2006; Tomlinson et al., 2011; Abir et al., 2021; Mahata et al., 2024). The UHI effect is influenced by factors such as heat waves, psychometric conditions, land surface changes, and light intensity (Kafy et al., 2021a, 2021b). The UTFVI is calculated as:

$$UTFVI = \frac{LST - LST_{\text{mean}}}{LST_{\text{mean}}} \quad (15)$$

3.8. Correlation analysis

Pearson’s correlation coefficient (r) method was applied to examine the relationships between land surface temperature (LST) and various spectral indices, including NDVI, NDBI, and NDWI, providing deeper insights into their interconnections (Kafy et al., 2021a, 2021b; Roustae et al., 2018; Mahata et al., 2024). The correlational data were derived via the “Create Fishnet” and “Extract Multiple Values to Point” tools in ArcGIS 10.8 software (Haldar et al., 2023). Furthermore, Pearson correlation analysis was used to improve the understanding of the relationships between various spectral indicators and LST. Person’s correlation was based on Eq. (16) as described by Obilor and Amadi (2018) and Saha et al. (2024).

$$pr = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \times \sum (y_i - \bar{y})^2}} \quad (16)$$

4. Results

In accordance with the methodology described in Section 3, this section presents the results derived from multitemporal satellite imagery, including analyses of LULC, LST, NDVI, NDBI, NDWI, UHI, and UTFVI. A comprehensive descriptive interpretation of these findings is provided below.

4.1. Assessing land use and land cover dynamics (1999–2024)

The study area was divided into four types of land cover, including water body, vegetation, urban areas, and bare land (Table 3).

Table 3
Changes in LULC types from 1999 to 2024.

LULC Type	1999		2011		2024		Change Rate in LULC Class		
	Area (km ²)	Percentage (%)	Area (km ²)	Percentage (%)	Area (km ²)	Percentage (%)	1999 to 2011	2011 to 2024	1999 to 2024
Waterbody	15.5	3.4	9.5	2.1	8.5	1.9	−39 %	−10 %	−45 %
Vegetation	287.6	62.8	278.5	60.8	262.3	57.3	−3 %	−6 %	−9 %
Urban Area	61.3	13.4	66.9	14.6	118.3	25.8	+9 %	+77 %	+93 %
Bare Land	93.6	20.4	103.1	22.5	68.9	15.1	+10 %	−33 %	−26 %

Fig. 3 shows the spatial distribution of LULC for the years between 1999 and 2024, maintaining decadal intervals.

Water bodies, vegetation, and bare land revealed a gradual decreasing trend, whereas urban areas showed an increasing trend (Fig. 4). The area covered by water bodies was 3.4 % (15.5 km²) in 1999, 2.1 % (9.5 km²) in 2011, and 1.9 % (8.5 km²) in 2024. Similarly, vegetation significantly declined over time, accounting for 62.8 % (287.6 km²) in 1999, 60.8 % (278.5 km²) in 2011, and 57.3 % (262.3 km²) in 2024. This decline is attributed primarily to unregulated development, land encroachment, and insufficient conservation of wetlands and ponds. Additionally, the area of bare land depicted a slightly decreased trend from 20.4 % (93.6 km²) to 15.1 % (68.9 km²) from 1999 to 2024. This decline is primarily due to the rapid expansion of settlements, agricultural activities and industrial infrastructure in this region. The Urban area expanded substantially, increasing from 13.4 % (61.3 km²) in 1999 to 14.6 % (66.9 km²) in 2011, and then rapidly to 25.8 % (118.3 km²) in 2024. This represents an overall growth of approximately 93 % over the 25-year period, underscoring the intensifying pace of urbanization and built-up land conversion. Most of this growth occurred between 2011 and 2024, when the urban footprint expanded by 77 %. This rapid urban growth is closely linked to industrialization, population influx, and infrastructure development, especially around the Dhaka-Gazipur corridor.

Over the past 25 years, the share of urban areas has nearly doubled their share (13.4 % to 25.8 %), reflecting rapid urbanization and land conversion. These trends highlight the interplay of urban expansion, vegetative loss, and environmental pressures within the study area.

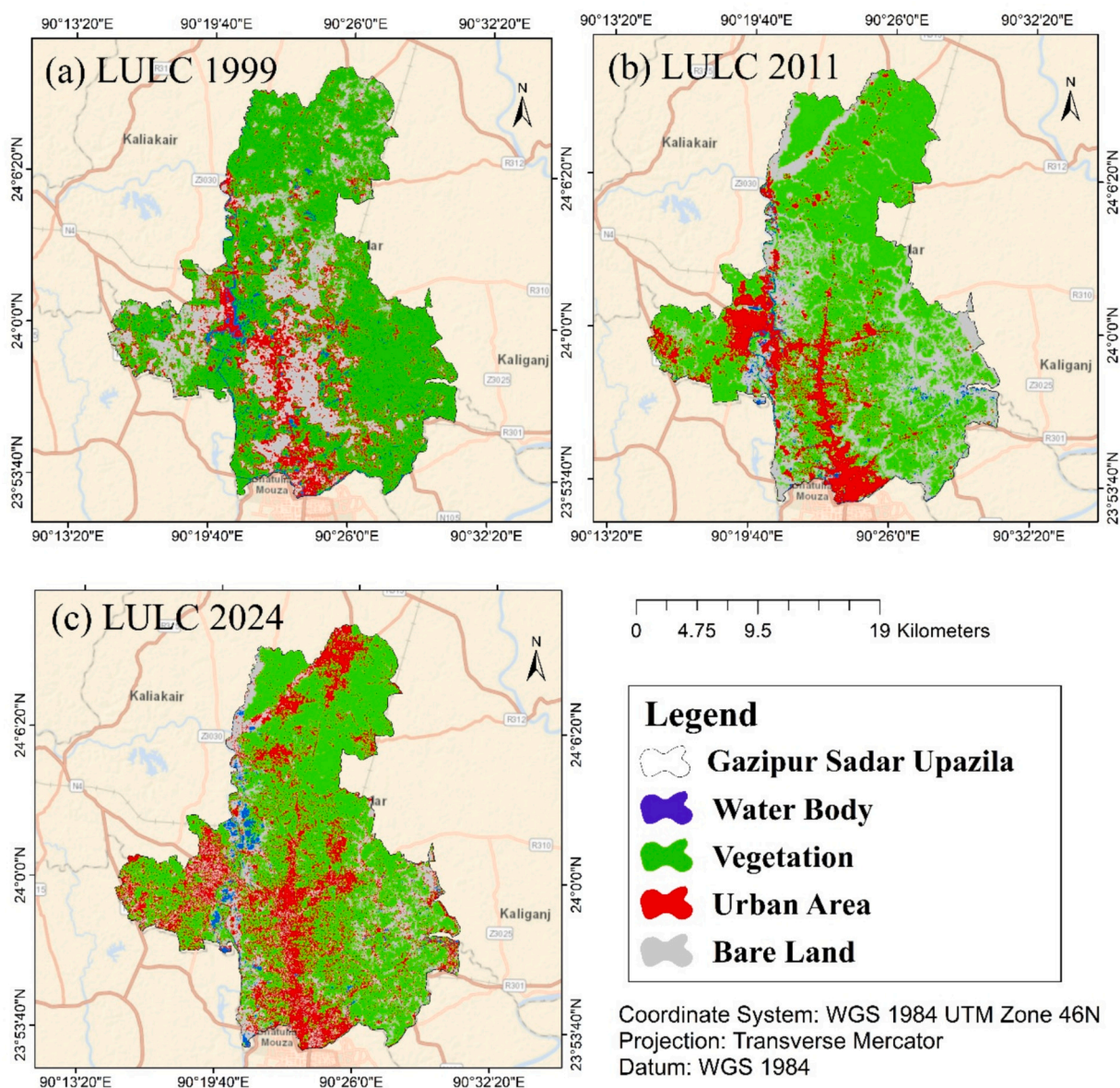


Fig. 3. Land use/land cover maps (a) 1999, (b) 2011, and (c) 2024.

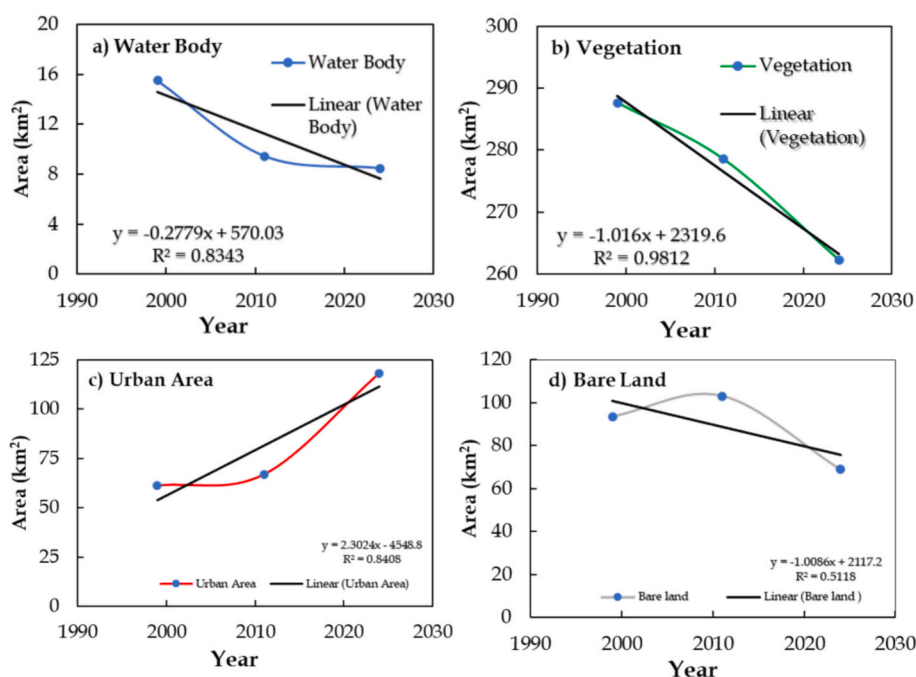


Fig. 4. Trend lines in different land use classes (a) water body, (b) vegetation, (c) urban area, and (d) bare land.

4.2. Accuracy assessment

Table 4 presents the accuracy assessment results for all study years, which delineate the user, producer, overall accuracy, and kappa coefficient. The results indicate that the classification is generally accurate, albeit with some discrepancies between user and producer accuracy. For example, in 1999, the user accuracy for waterbodies was 75 %, whereas the producer accuracy was 66 %. This signifies that while 66 % of actual waterbody areas were correctly classified, 25 % of the waterbody pixels in the classified image were erroneously labeled as other land cover types. In 2024, both user and producer accuracy of vegetation and urban areas achieved 100 % accuracy, demonstrating perfect classification. However, waterbody classification still exhibited a producer accuracy of 75 %, indicating that some areas of water bodies were misclassified as other land cover types. As shown in Table 4, the overall accuracy improved over the time, increasing from 86 % in 1999 to 94 % in 2024. This improvement suggests an enhancement in classification methodologies over time. The kappa coefficient also increased, from 79 % in 1999 to 92 % in 2024, indicating a higher reliability of the classification. The lower kappa value in 1999 suggests a higher incidence of errors in the classification, particularly for water bodies and urban areas. By 2024, the classification demonstrated significantly fewer errors, attributable to improved techniques and data quality.

Furthermore, Google Earth Pro was used to develop reference maps for accuracy assessment, supporting the validation of the classification outputs. However, the 1999 imagery obtained from the USGS Earth Explorer had limitations in both spatial resolution and historical coverage. These limitations particularly affect the accurate identification of certain land cover types. Therefore, these LULC classes had relatively lower user and producer accuracies in 1999 than in later years. Despite these challenges, the overall classification performance was acceptable, and the resulting LULC maps are considered reliable for assessing long-term land cover changes.

4.3. Spatiotemporal LULC changes (1999–2024)

The changes in different LULC types of Gazipur Sadar Upazila from 1999 to 2024 are shown in Fig. 5. Significant landscape changes

Table 4

Accuracy assessment of land use and land cover classes for 1999, 2011, and 2024.

Year	User accuracy (%)				Producer accuracy (%)				Overall Accuracy	Kappa coefficient (%)
	Waterbody	Vegetation	Urban Area	Bare Land	Waterbody	Vegetation	Urban Area	Bare Land		
1999	75	100	85	100	66	100	85	100	86	79
2011	82	90	95	80	90	80	87	88	87	83
2024	95	100	100	86	75	100	100	85	94	92

have occurred over the last two decades due to urban growth and land development. The mapped data highlight a large increase in urban areas, mainly in the central and southern regions. Between 1999 and 2024, urban areas in the study region expanded by 57.0 km², representing a significant increase of approximately 93 %, as illustrated in Table 3. A major trend is the conversion of vegetation into urban land, which indicates deforestation and the rising need for buildings and infrastructure. Vegetation also showed a declining trend, with the highest loss of 16.2 km² (−6 %) occurring from 2011 to 2024. Additionally, a large portion of bare land has turned into built-up areas, resulting in rapid urban expansion. A steady increase in urban areas occurred in each period, with the highest growth of +51.4 km² (+77 %) observed between 2011 and 2024. Water bodies have also changed, with some areas becoming urban settlements, which may affect local water systems and ecosystems. Water bodies showed a continuous decline, with the largest decrease of 6.1 km² (−39 %) occurring between 1999 and 2011. However, some vegetation zones remain unchanged, indicating some level of environmental stability. (See Fig. 6.)

4.4. Spatiotemporal changes in land surface temperature from 1999 to 2024

The seasonal land surface temperature (LST) was analyzed via thermal band data from satellite images collected between 1999 and 2024. LST changes from 1999 to 2024 were analyzed using Landsat thermal bands, focusing on both winter and summer seasons to examine seasonal thermal dynamics. In Fig. 7, red indicates areas with higher LST, while blue represents cooler zones, effectively visualizing the spatial variability of surface temperature in the study area. This approach provides a detailed understanding of seasonal variations in temperature over time. Seasonal variations in LST during winter and summer from 1999 to 2024 are presented in Table 5.

4.4.1. Land surface temperature (LST) in winter

Fig. 7 (a, b, and c) shows the LST distribution of the study area in the winter season from 1999 to 2024. The LST during the winter season has noticeably changed over the years 1999, 2011, and 2024 (Table 5).

In 1999, the highest recorded temperature was 28.5 °C. This value decreased to 27.0 °C in 2011 and then slightly increased to 27.1 °C in 2024, indicating a small fluctuation over time. The lowest temperature recorded during winter also varied significantly. In 1999, the temperature was 17.0 °C, but it decreased to 12.2 °C in 2011, indicating a significant decrease. However, by 2024, it has increased again to 16.7 °C. This could be due to changes in land use, vegetation cover, and urban expansion, which can influence the heat retention and cooling processes of the landscape.

Similarly, the average winter temperature followed a changing trend. In 1999, the mean temperature was recorded at 21.7 °C. This value decreased to 18.7 °C in 2011, indicating a colder winter season. However, in 2024, the mean temperature increased again to 19.2 °C. The variation in temperature within each winter season, represented by the standard deviation, was also different for each year. Winter temperature variability increased significantly over the study period, as shown by the increasing standard deviation from 0.97 °C (1999) to 1.20 °C (2024). This trend is further supported by the increasing coefficient of variation, which increased from 4.47

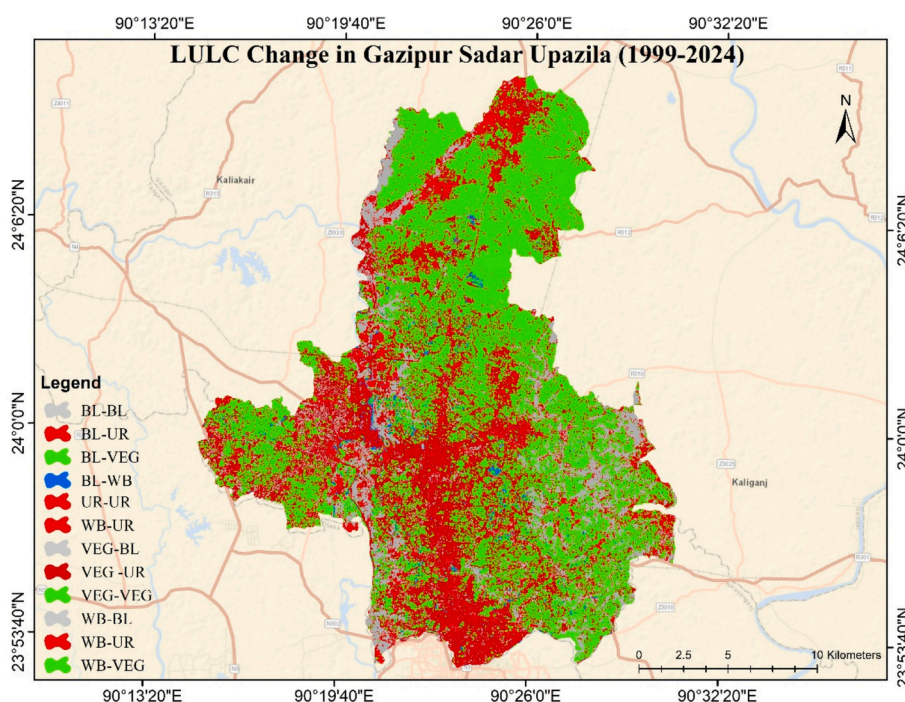


Fig. 5. LULC conversion between 1999 and 2024 in Gazipur Sadar Upazila.
Abbreviations: UR – Urban area, VEG – Vegetation, BL – Bare Land, WB – Water Body.

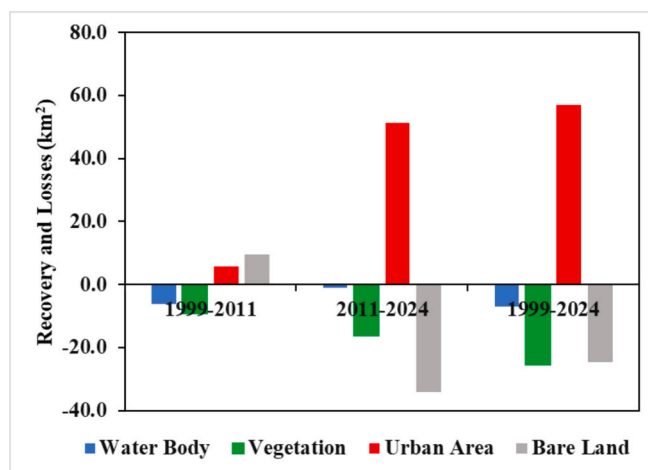


Fig. 6. Recovery and losses of land cover classes from 1999 to 2024.

% to 6.25 %, demonstrating greater interannual fluctuations in winter temperatures.

These trends suggest that winter temperatures have not remained constant over time. The year 2011 experienced the coldest winter, with the lowest mean and minimum temperatures. In contrast, 2024 showed a slight recovery, with temperatures increasing again but not reaching the highs of 1999.

4.4.2. Land surface temperature (LST) in summer

Fig. 7 (d, e, and f) shows the LST variations in the summer season from 1999 to 2024. The LST values ranged from 34.1 °C in 1999, increased to 38.4 °C in 2011, and further increased to 39 °C in 2024. The minimum temperatures for the same periods were 21.5 °C in 1999, 23.7 °C in 2011, and 26 °C in 2024 (Table 5). The consistent increase in summer LST indicates an increasing trend, which could be attributed to the conversion of vegetation cover into urban areas and bare land. The most intense heat zones became more spatially extensive, particularly in the southern and southwestern urban-industrial zones.

The mean LST in summer followed a similar increasing pattern, rising from 26.8 °C in 1999 to 27.1 °C in 2011, and reaching 32.5 °C in 2024. The SD values for the summer LST also varied, measuring 1.93 °C in 1999, and 1.39 °C in 2011, and slightly decreased to 1.61 °C in 2024. These variations suggest that dynamic changes in land surface properties are driven primarily by urbanization and deforestation. The sharp increase in the maximum LST values from 1999 to 2024 is particularly concerning, as it reflects the impacts of declining vegetation, loss of water bodies, and unregulated urban sprawl.

Overall, the maps show that urban expansion and land cover change especially the loss of vegetation and increase in impervious surfaces significantly contributed to increased LSTs across both seasons. The most affected zones correspond with areas of dense construction and industrial growth in the region.

4.5. Relationships between LULC and LST

This study investigates the interrelationship and examines how land use/land cover (LULC) affects the LST in Gazipur Sadar Upazila. LST and LULC data from 1999, 2011, and 2024 were analyzed for both the summer and winter seasons to assess variations in LST across different LULC classes. The results revealed variations in temperature ranges across different land types, including water bodies, vegetation, urban areas, and bare land during the winter and summer seasons (Supplementary Figs. 1 and 2).

4.5.1. Winter LST dynamics and LULC classes

Water bodies show a decline in maximum temperature in the winter seasons, ranging from 25.0 °C in 1999 to 22.1 °C in 2011 and 21.5 °C in 2024, whereas minimum temperatures also decline from 18.3 °C in 1999 to 16.7 °C in 2024. The maximum LST in the vegetation areas increased from 27.0 °C in 1999 to 23.0 °C in 2011. However, it slightly increased to 24.4 °C in 2024. Urban areas and bare land consistently presented the highest mean LST values (Table 6). The maximum LST in the urban areas slightly decreased from 28.5 °C in 1999 to 27.0 °C in 2011 and remained the same in 2024. Bare land showed more variation. The maximum LST decreased from 27.2 °C in 1999 to 21.2 °C in 2011 but increased to 27.1 °C in 2024. Fig. 8 reveals that urban and bare land areas presented significantly higher LST compared to vegetation and water bodies. These findings strengthen the association between urban expansion and increased surface temperatures in the region.

4.5.2. Summer LST dynamics and LULC classes

In the summer seasons, water bodies exhibit maximum temperatures increasing from 30.0 °C in 1999 to 33.9 °C in 2024, with minimum temperatures also increasing from 22.4 °C to 25.9 °C during the same period. The vegetation areas show a similar trend, with maximum LST increasing from 32.1 °C in 1999 to 35.8 °C in 2024, and the minimum temperature increasing from 21.5 °C to 25.9 °C.

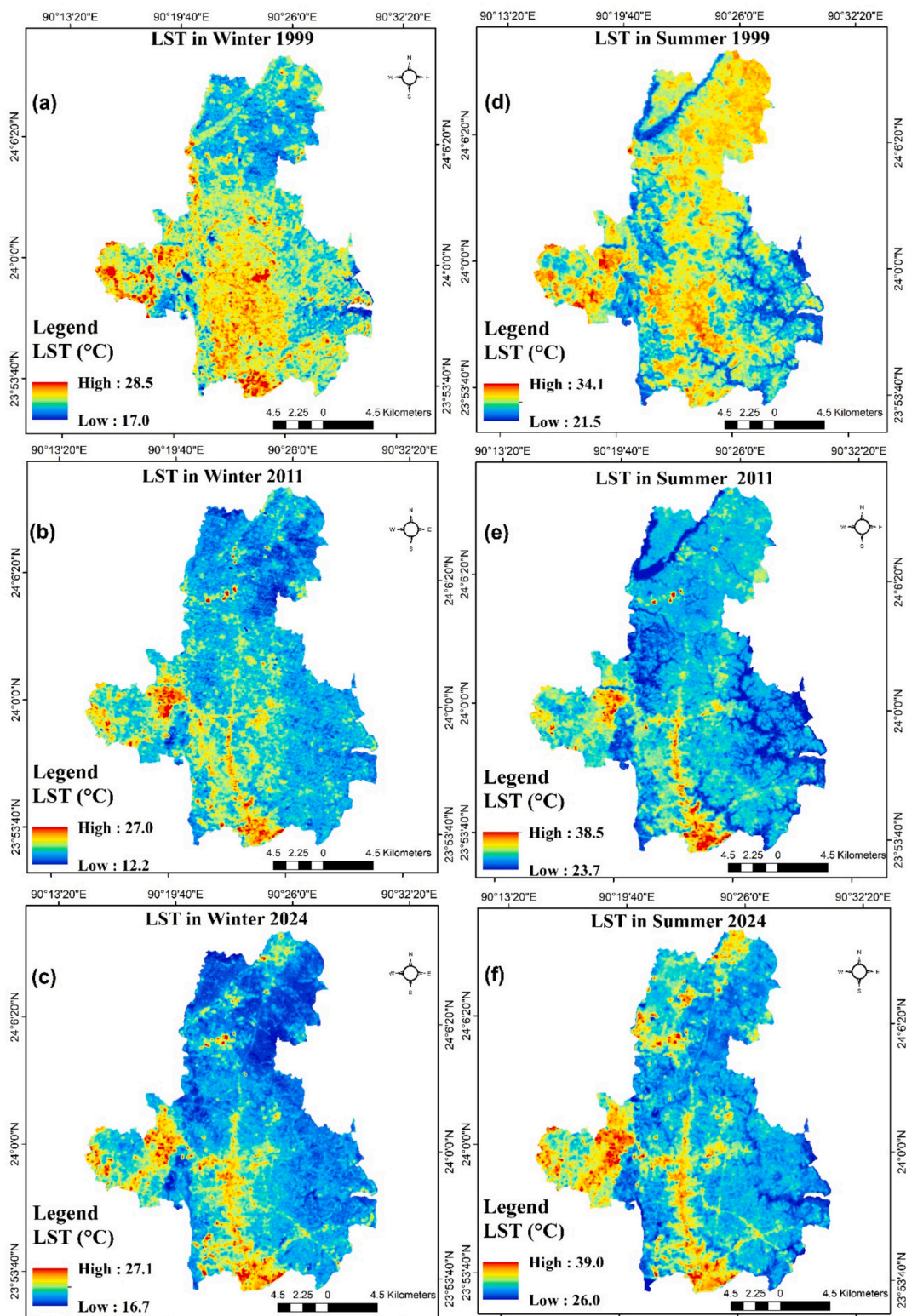


Fig. 7. Land surface temperature maps in winter (a, b, and c), summer (d, e, and f) in 1999, 2011, and 2024.

Table 5
Seasonal variations in LST in winter and summer from 1999 to 2024.

LST in Winter	1999	2011	2024
Max (°C)	28.5	27.0	27.1
Min (°C)	17.0	12.2	16.7
Mean (°C)	21.7	18.7	19.2
SD (°C)	0.97	1.04	1.20
CV (%)	4.47	5.57	6.25
LST in Summer	1999	2011	2024
Max (°C)	34.1	38.4	39.0
Min (°C)	21.5	23.7	26
Mean (°C)	26.8	27.1	32.5
SD (°C)	1.93	1.39	1.61
CV (%)	7.20	5.13	4.95

Urban areas experience the most significant temperature fluctuations in the summer, with maximum temperatures increasing from 34.1 °C in 1999 to 39 °C in 2024, and minimum temperatures rising from 23.7 °C to 26.0 °C (Fig. 9). Urban areas have the highest mean temperature during the summer season. These changes reflect the increasing dominance of impervious surfaces and a reduction in natural cooling mechanisms, driven by unplanned urbanization and industrial growth. Bare land also significantly increased, with the maximum LST increasing from 33.3 °C in 1999 to 38.8 °C in 2024, whereas the mean temperatures increase from 28.8 °C to 29.6 °C (Table 6).

4.6. Spatiotemporal dynamics of NDVI, NDBI, and NDWI from 1999 to 2024

Fig. 12 shows changes in vegetation, built-up areas, and water bodies from 1999 to 2024. We use three indices: normalized difference vegetation index (NDVI), normalized difference built-up index (NDBI), and normalized difference water index (NDWI). These indices help us understand how land is changing over time.

In 1999, the highest NDVI value was 0.69, but by 2011, it had decreased to 0.59. In 2024, it declined further to 0.41 (Fig. 10 (a, b, and c)). This shows a clear reduction in green areas, likely due to deforestation and urban growth. The lowest NDVI values also increased from −0.36 in 1999 to −0.10 in 2024, suggesting that previously vegetated or water-covered areas have been replaced by non-vegetated surfaces, leading to an overall loss of vegetation. The NDBI maps show increased built-up areas (Fig. 10 (d, e, and f)). In 1999, the NDBI value was 0.27, but by 2011, it had increased to 0.42. This suggests a rapid rise in urbanization. However, in 2024, the highest NDBI value increased to 0.61. The rising NDBI suggests accelerated urban expansion. The lowest NDBI values also changed significantly, decreasing from −0.33 in 1999 to −0.63 in 2024, indicating a growing distinction between urban and non-urban surfaces, possibly due to the conversion of vegetated or bare land into dense built-up areas. The NDWI maps (Fig. 10 (g, h, and i)) reveal that water bodies are decreasing. In 1999, the highest NDWI value was 0.32, which increased significantly to 0.35 in 2011. However, by 2024, the highest NDWI value declined substantially to 0.13, suggesting a significant reduction in water bodies. The lowest NDWI values also changed from −0.62 in 1999 to −0.36 in 2024. This decline suggests that lakes, rivers, and other water sources are drying.

Overall, from 1999 to 2024, Gazipur Sadar Upazila faced major environmental changes. The normalized difference vegetation index (NDVI), which measures plant health and growth, decreased by 40.6 % (from 0.69 to 0.41). This shows a severe loss of healthy green areas. Moreover, built-up areas have increased rapidly, and water bodies have decreased significantly. These changes cause severe damage to the environment. To protect the region, better planning is needed to balance city growth with saving nature and water resources.

4.7. Correlations between different spectral indices and LST

The Pearson correlation coefficients between the LST and spectral indices were analyzed via Origin Pro, Version 2023 (Table 7). In 1999, the LST had a strong negative correlation with the NDVI ($r = -0.79$), indicating that areas with more vegetation tended to be cooler. There is a moderate positive moderate correlation between the LST and NDBI ($r = 0.58$), suggesting that urban areas, which have more buildings, tend to be warmer. The LST also has a strong positive correlation with NDWI ($r = 0.72$), indicating that water bodies help cool the land surface. By 2011, the correlation between the LST and NDVI weakened ($r = -0.035$), indicating that vegetation had less of an effect on surface cooling, possibly because of increased urban development. The relationship between the LST and NDBI becomes stronger ($r = 0.60$), indicating that urban areas have an even greater impact on rising temperatures. The LST-NDWI correlation also becomes slightly negative ($r = -0.059$), suggesting that water bodies play a smaller role in cooling the area.

In 2024, the correlation between the LST and NDBI is strongly positive ($r = 0.85$), revealing a clear link between urban growth and higher temperatures. The LST-NDVI correlation is still negative ($r = -0.23$), although it is weaker than before. The correlation between the LST and NDWI is very weak ($r = 0.13$), meaning that water bodies have almost no impact on the LST by this time.

Over the years, the NDVI and NDWI exhibited strong negative correlations of −0.97, −0.93, and −0.98 in 1999, 2011, and 2024, respectively. This suggests that as vegetation increases, water availability decreases, particularly as areas covered by water are taken over by vegetation or urban land.

Overall, the positive correlation between the LST and NDBI further supports this observation. In 2024, the correlation coefficient

Table 6

	Winter- 1999			Winter- 2011			Winter- 2024			Change of Mean Temperature (°C)		
Land type	Max. (°C)	Min. (°C)	Mean (°C)	Max. (°C)	Min. (°C)	Mean (°C)	Max. (°C)	Min. (°C)	Mean (°C)	(1999–2011) (°C)	(2011–2024) (°C)	(1999–2024) (°C)
Water Body	25.0	18.3	21.6	22.1	16.5	18.4	21.5	16.7	17.9	−3.2	−0.5	−3.7
Vegetation	27.0	18.3	21.3	23.0	13.7	18.6	24.4	16.8	20.6	−2.7	2.0	−0.7
Urban Area	28.5	16.9	22.7	27.0	12.2	20.2	27.0	16.6	21.8	−2.5	1.6	−0.9
Bare land	27.2	18.8	22.3	21.2	16.5	18.2	27.1	16.9	22.0	−4.1	3.8	−0.3
	Summer-1999			Summer-2011			Summer-2024			Change of Mean Temperature (°C)		
Land Type	Max. (°C)	Min (°C)	Mean (°C)	Max (°C)	Min. (°C)	Mean (°C)	Max. (°C)	Min. (°C)	Mean (°C)	(1999–2011) (°C)	(2011–2024) (°C)	(1999–2024) (°C)
Water Body	30.0	22.4	25.6	30.4	24.1	26.5	33.9	25.9	28.1	0.9	1.6	2.5
Vegetation	32.1	21.5	26.0	32.5	24.1	27.2	35.8	25.9	29.3	1.2	2.1	3.3
Urban Area	34.1	23.7	29.0	38.4	24.1	31.3	39.0	26.0	32.5	2.3	1.2	3.5
Bare Land	33.3	25.8	28.8	34.0	23.7	28.9	38.8	25.9	29.6	0.1	0.7	0.8

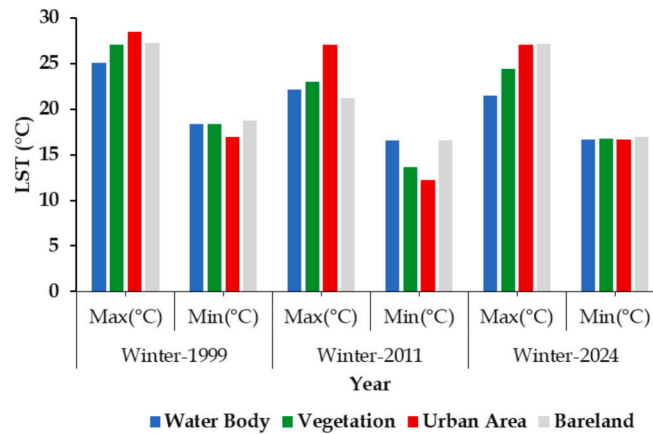


Fig. 8. LST (Max. and Min.) profile of different LULC types in winter.

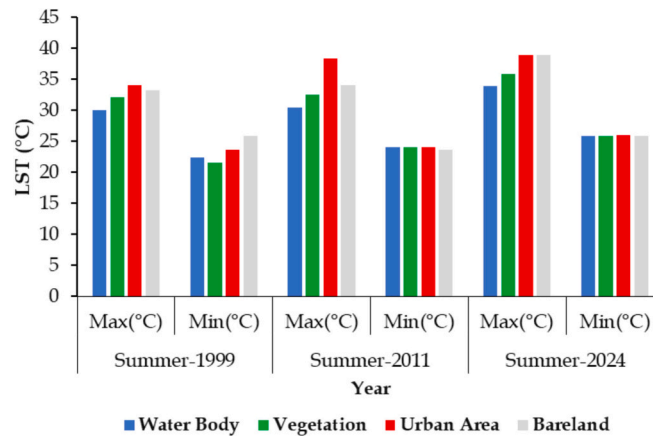


Fig. 9. LST (Max. and Min.) profile of different LULC types in summer.

was 0.85, indicating that the NDBI and LST had strong positive relationships. This indicates that surface temperatures rose in parallel with the NDBI. In all years, there was a positive relationship between the LST and the NDBI. This means that as urbanization increased, surface temperatures also rose. Our research revealed that urban growth, deforestation, and water body decline are major factors influencing increasing urban land surface temperatures.

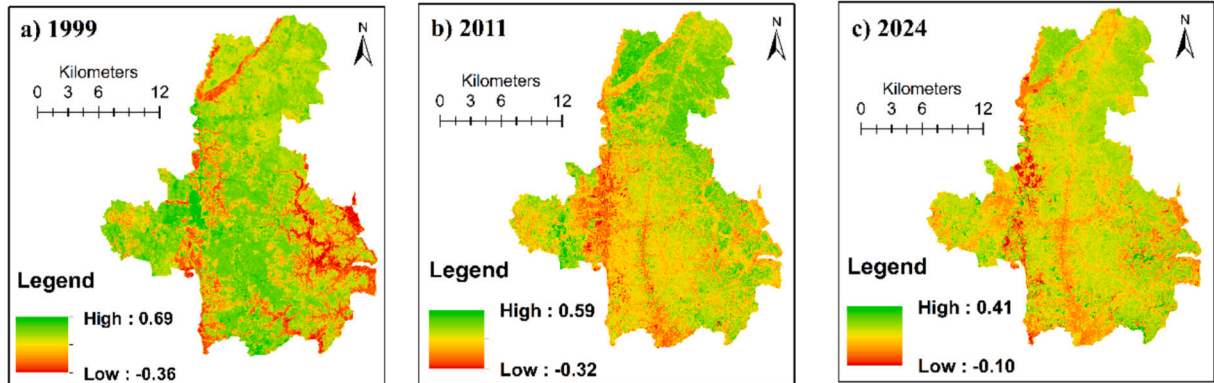
4.8. Regression model

A simple linear regression model (SLRM) was applied to assess the influence of key land surface indicators (NDBI and NDVI) on land surface temperature across selected years (Fig. 11). LST served as the dependent variable. Moreover, either the NDVI or the NDBI functioned as independent variables. The regression equations and associated coefficients shown in the relevant figures indicate the strength and direction of the relationships, highlighting how changes in vegetation and built-up areas affect surface temperature. The regression analysis for 1999 revealed that the NDBI accounted for 34 % of the variation in LST ($R^2 = 0.3408$), indicating a moderate positive association. Moreover, the NDVI explained 62 % of the variation ($R^2 = 0.6279$), reflecting a significant cooling effect. The relationship weakened in 2011, with the NDBI accounting for 36.5 % of the LST variation ($R^2 = 0.3645$), whereas the NDVI exhibited an extremely weak association with the LST, explaining 0.1 % of the variation ($R^2 = 0.0012$), indicating an almost negligible cooling effect during this year. By 2024, NDBI maintained a strong positive relationship with the LST, explaining 72 % of its variation ($R^2 = 0.7204$), suggesting the continued influence of urbanization on thermal patterns. However, the NDVI accounted for merely 5.2 % of the LST variation ($R^2 = 0.0518$), reflecting a diminished vegetative cooling effect in more recent years.

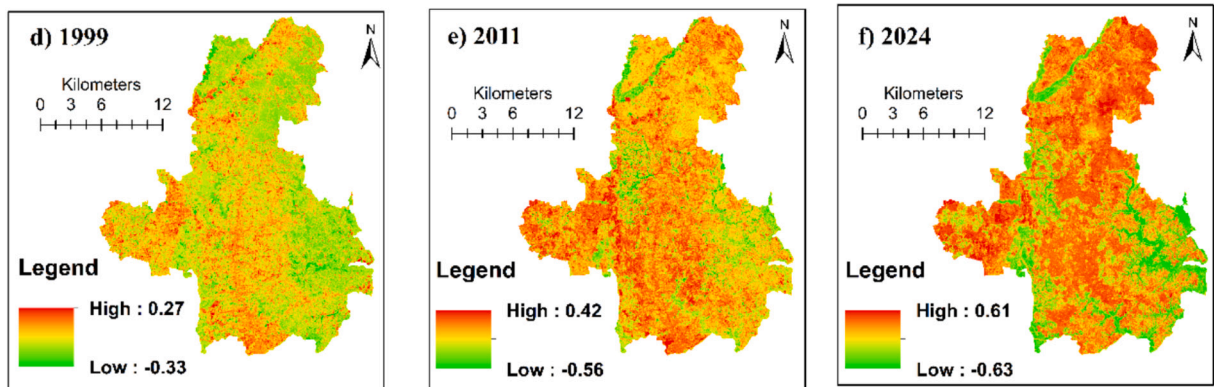
4.9. UHI and UTFVI analysis

This study used the urban thermal field variance index (UTFVI) to measure how urban heat island (UHI) affects ecological degradation. The maps show changes in urban heat island (UHI) intensity and urban thermal field variance index (UTFVI) from 1999 to

Normalized Difference Vegetation Index (NDVI)



Normalized Difference Built-up Index (NDBI)



Normalized Difference Water Index (NDWI)

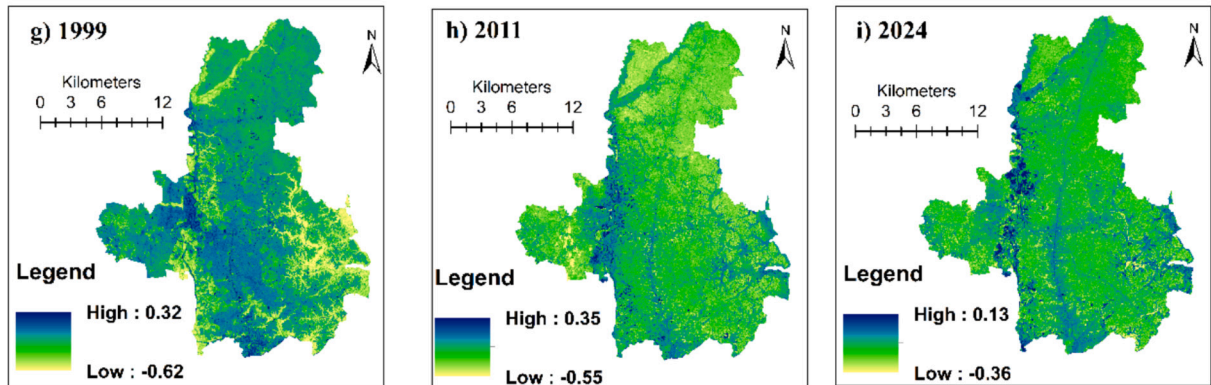


Fig. 10. Spatiotemporal analysis of NDVI (a, b, and c), NDBI (d, e, and f), and NDWI (g, h, and i) from 1999 to 2024.

2024 due to urbanization (Fig. 12). In 1999 (Fig. 12 (a, d)), red zones were small and scattered, mainly in the central and southwestern areas, whereas blue zones were widespread in the north and east, indicating stable ecological conditions with extensive green areas and fewer heat-affected zones. By 2011 (Fig. 12 (b, e)), urbanization expanded the red zones, particularly in the central, southwestern, and southern regions, whereas blue zones shrank as natural land decreased, leading to increased urban heat stress and an imbalance in the ecological system. By 2024 (Fig. 12 (c, f)), red zones had spread further into the west and north, increasing heat risk. The central and western areas now have more red zones due to continuous urban growth, whereas cooler zones have become less prominent. In terms of the ecological evaluation indices (Table 8), according to the UTFVI values, the ecological conditions have worsened, with increasing temperature stress, habitat loss, and reduced vegetation cover. There are fewer blue zones, mostly in the east and southeast, where vegetation helps cool the area, however overall, the environment faces significant heat stress and ecological degradation in

Table 7
Correlations between difference spectral indicators (including the NDVI, NDBI, and NDWI) and the LST in (a) 1999, (b) 2011, and (c) 2024.

(a) 1999				
	LST	NDVI	NDBI	NDWI
LST	1	−0.79	0.58	0.72
NDVI	−0.79	1	−0.88	−0.97
NDBI	0.58	−0.88	1	0.79
NDWI	0.71	−0.97	0.79	1
(b) 2011				
	LST	NDVI	NDBI	NDWI
LST	1	−0.035	0.60	−0.059
NDVI	−0.035	1	0.077	−0.93
NDBI	0.60	0.077	1	−0.25
NDWI	−0.059	−0.93	−0.25	1
(c) 2024				
	LST	NDVI	NDBI	NDWI
LST	1	−0.23	0.85	0.13
NDVI	−0.23	1	−0.27	−0.98
NDBI	0.85	−0.27	1	0.12
NDWI	0.13	−0.98	0.12	1

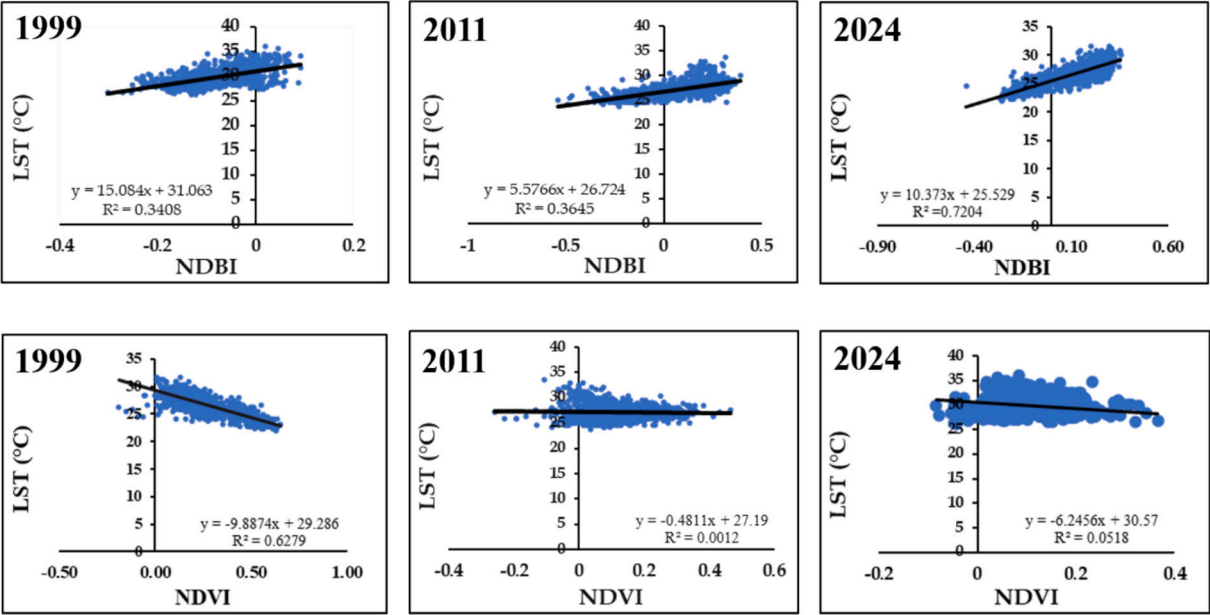


Fig. 11. Correlations between the LST, NDBI, and NDVI for 1999, 2011, and 2024.

urban areas.

5. Discussion

This study investigated the seasonal patterns of land surface temperature (LST), land use and land cover (LULC) changes, and their environmental connections in Gazipur Sadar Upazila, Bangladesh, over 25 years from 1999 to 2024. The findings show that urban areas have expanded rapidly, especially in recent years, resulting in major transformations in land use and a noticeable rise in surface temperatures.

The percentage of urban land increased from 13.4 % to 25.8 %, showing 93 % growth. This expansion mostly replaced vegetated areas and water bodies, which decreased by 9 % and 45 %, respectively. These patterns are similar to the observations of [Rakib et al. \(2020\)](#), [Roy et al. \(2021\)](#), [Abir and Saha \(2021\)](#) and [Saha et al. \(2024\)](#), who reported rapid urban growth and ecological change in urban regions of Bangladesh. Dhaka city has the fastest urban growth in Bangladesh, and its close location has contributed to the rapid

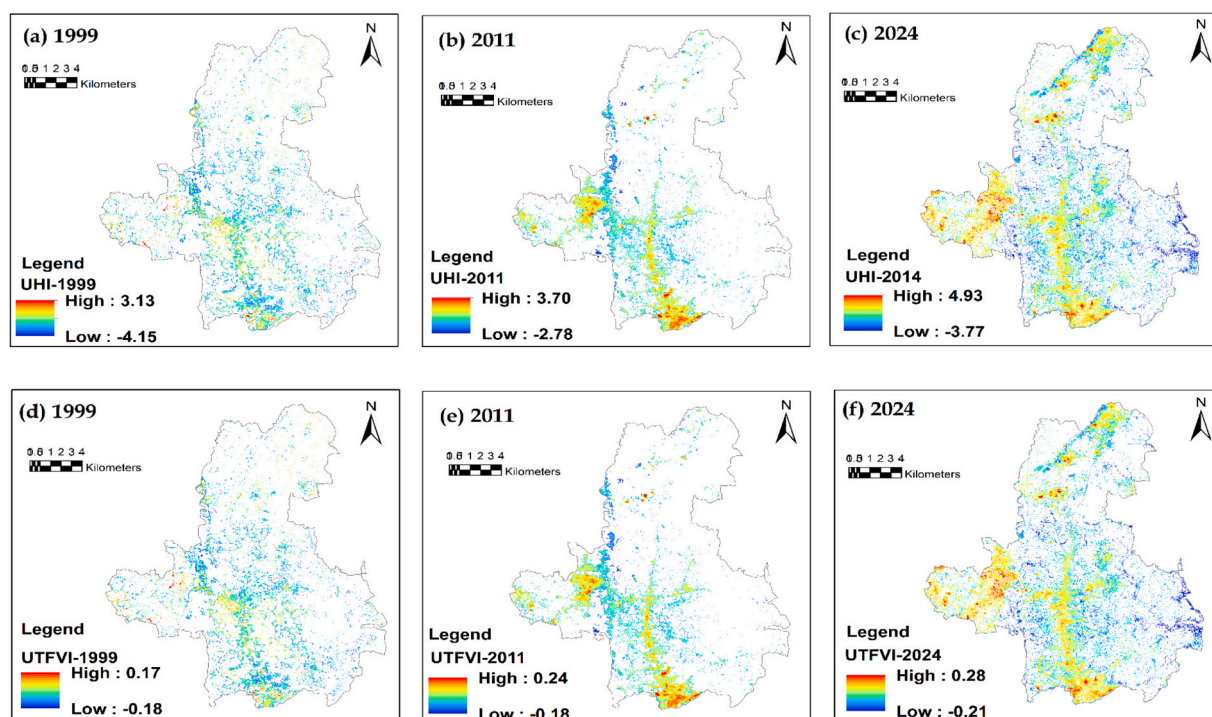


Fig. 12. Spatiotemporal dynamics of the Urban Heat Island (UHI) effect and Urban Thermal Field Variance Index (UTFVI) in Gazipur Sadar Upazila for the years 1999, 2011, and 2024.

Table 8

Pattern of urban thermal field variance index scale (Zhang et al., 2006).

UTFVI Value	UHI	Ecological Evaluation Index
<0.000	None	Excellent
0.000–0.005	Middle	Normal
0.010–0.015	Strong	Bad
0.015–0.020	Stronger	Worse
>0.020	Strongest	Worst

urban growth of Gazipur Sadar Upazila (Roy et al., 2021). The rapid urban expansion in Gazipur Sadar, similar to Pabna's 358 % built-up growth reported by Abir and Saha (2021), reflects a national trend of urban sprawl replacing vegetation and water bodies, driving surface temperature increases and ecological stress in the urban areas of Bangladesh. The expansion of urban areas and the degradation of natural resources result from extensive urban development, rapid population growth in cities, rural-to-urban migration, and the region's strategic and economic significance (Weng et al., 2004; Naim and Kafy, 2021; Abir and Saha, 2021; Hasnat, 2022; Roy et al., 2021).

The analysis of LST revealed a clear increase in summer temperatures, increasing from an average of 26.8 °C in 1999 to 32.5 °C in 2024. Changes in land surface temperature (LST) result from complex geographical processes, where rapid population growth and increased development activities contribute to increased LST (Chen et al., 2021; Kafy et al., 2021a, 2021b). Several factors contribute to changes in land surface temperature (LST), often working together due to complex geographical processes, with rapid population growth and increased development activities jointly driving the rise in LST (Chen et al., 2021; Kafy et al., 2020). Previous studies have demonstrated that reducing urban green spaces can increase land surface temperatures by approximately 2–5 °C, primarily through decreased shading and limited evapotranspiration (Jandaghian and Colombo, 2024).

According to zonal classification observations of LULC and LST in winter and summer, urban areas and bare lands were warmer than the mean LST, and water bodies and vegetation remained relatively cooler (Table 6). These results are consistent with earlier studies by Choudhury et al. (2019), Santamouris (2015), and Imran et al. (2021), who reported that built-up areas tend to trap more heat due to reduced greenery and more heat-absorbing surfaces. Many researchers have also reported that water bodies and vegetation are associated with low LST (Gazi et al., 2021; Guha and Govil, 2022; Mahata et al., 2024; Saha et al., 2021). The Intergovernmental Panel on Climate Change (IPCC) forecasts that by 2100, global average surface temperatures will rise between 1.4 °C and 5.8 °C, with atmospheric carbon dioxide levels increasing to double those recorded during pre-industrial times (Singh et al., 2017).

Environmental indices were used to study the relationships between surface temperature and land cover types. The LST was

positively correlated with the NDBI (Table 6), explaining 72 % of the LST variability in the regression model, indicating that more built-up areas contributed to higher temperatures. However, the NDVI exhibited a negative correlation with LST, indicating that areas with more vegetation tended to be cooler. Compared to other seasons, NDVI has the weakest relationship with the LST during winter (Guha et al., 2020). The regression model revealed that the NDVI accounted for 0.1 % to 62 % of LST variability. These relationships support findings by Edan et al. (2021), Hasnat (2022), Patel et al. (2024), Mahata et al. (2024), and Abir and Saha (2021) highlighting the usefulness of spectral indices in understanding urban heat dynamics. Vegetation significantly contributes to the regulation of urban thermal environments by providing shading and facilitating evapotranspiration, which directly influences local temperature levels and enhances overall thermal comfort (Zhao et al., 2025).

The UTFVI revealed that ecological stress has increased over time, especially in densely urbanized areas. A poor ecological status was observed in the high-density built-up zones, reflecting increased thermal pressure and environmental degradation. The findings of this study are consistent with those of Abir et al. (2021), who reported that urban areas experience more adverse thermal and ecological conditions compared to their surrounding non-urban regions in Bogura, Bangladesh. In Gazipur Sadar Upazila, a similar pattern was observed, where urban zones exhibited significantly elevated land surface temperatures and increased ecological stress relative to adjacent rural areas. This result is also in line with similar findings by Mahata et al. (2024) in New Town, Kolkata, where high levels of urban development also led to poor ecological conditions. Urban heat islands have become more intense and widespread, indicating a rise in thermal discomfort and reduced ecological quality. These results also agree with studies by Vohra et al. (2024) and Miah et al. (2024) who linked rising UTFVI values with worsening environmental conditions in urban centers.

6. Limitations and future work

This study has several limitations. The primary constraint was the reliance on cloud-free Landsat satellite data, which limited image availability, particularly in the monsoon-prone tropical region of Bangladesh. Additionally, since Landsat takes images every 16 days, it is difficult to obtain a full average LST for winter and summer (Abir and Saha, 2021). Differences in the dates of image collection may also affect the results. Manual interpretation of imagery introduces a potential source of human error, especially in distinguishing between visually similar classes such as vegetation and built-up areas. Furthermore, the 30-m spatial resolution of Landsat imagery has limited the ability to capture fine-scale land cover variations in heterogeneous urban environments. Methodologically, the study did not incorporate ground-truth or in-situ temperature data, which could have enhanced the accuracy of the LST estimations. These limitations suggest that future research should utilize higher-resolution datasets (e.g., Sentinel-2, MODIS) and integrate real-time environmental and socioeconomic data to provide a more comprehensive assessment of urban ecological dynamics. Future research should use higher-resolution satellite images and focus on predicting land use changes and their impact on the environment.

7. Conclusions

This study revealed that land use and land cover (LULC) in Gazipur Sadar Upazila changed significantly between 1999 and 2024. Water bodies and vegetation have decreased, mainly due to urban growth and industrial expansion. Urban areas have nearly doubled, replacing natural landscapes. These changes have led to environmental problems such as habitat loss, fewer green spaces, and shifts in local climate conditions. The study also revealed that land surface temperature (LST) has increased over time. The summer temperatures rose by 5.7 °C during the study period. The highest temperature increases were observed in urban and barren land areas, whereas vegetation and water bodies helped keep temperatures lower. The increase in minimum temperatures across all seasons suggests a long-term warming trend caused by urbanization and climate change. These findings confirm that urban growth has a strong effect on temperature, leading to the urban heat island (UHI) effect. Increasing vegetation and protecting water bodies can help reduce the LST. In this study, LST was positively correlated with the NDBI, indicating that the number of impervious surfaces increased, and the LST also increased. The LST and NDVI were negatively correlated, reflecting the cooling effect of vegetation on surface temperature. The negative correlation between the NDVI and NDBI highlights the significant impact of urbanization on reducing vegetation cover. Excellent ecological status was observed in areas with sparse and dense vegetation as well as water bodies, whereas poor ecological status was found in high-density built-up areas.

Overall, this research highlights the negative environmental effects of unplanned urban development in Gazipur Sadar Upazila. While urbanization has supported economic and infrastructure growth, it has also increased surface temperatures, reduced vegetation, and threatened the ecological balance. To reduce these impacts, sustainable urban planning should be promoted, including the development of green spaces, rooftop gardening, the restoring natural vegetation, and the protection of water bodies. These findings will help city planners and policymakers develop better use plans for a sustainable and green environment.

CRedit authorship contribution statement

Md. Rased Hasan Sojib: Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Md. Monabbir Hossain:** Writing – original draft, Visualization, Methodology, Formal analysis. **Md. Al-Amin Khan:** Writing – original draft, Methodology, Conceptualization. **Md. Abdul Baten:** Writing – review & editing, Supervision, Software, Resources. **Md. Bad-iuzzaman Khan:** Writing – review & editing, Supervision, Software, Resources, Investigation. **F.K. Sayema Tanzania:** Writing – review & editing, Resources, Data curation. **Mahmudul Hasan:** Writing – original draft, Data curation.

Funding statement

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declared that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

The authors would like to thank the USGS officials for providing the data required for the study free of cost. They are also grateful to the anonymous respondents who share their opinions during the collection of validation points.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.uclim.2025.102662>.

Data availability

Data will be made available on request.

References

- Abir, F.A., Saha, R., 2021. Assessment of land surface temperature and land cover variability during winter: A spatio-temporal analysis of Pabna municipality in Bangladesh. *Environ. Challen.* 4, 100167. <https://doi.org/10.1016/j.envc.2021.100167>.
- Abir, F.A., Ahmmed, S., Sarker, S.H., Fahim, A.U., 2021. Thermal and ecological assessment based on land surface temperature and quantifying multivariate controlling factors in Bogura, Bangladesh. *Heliyon* 7 (9), e08012. <https://doi.org/10.1016/j.heliyon.2021.e08012>.
- Ahmed, B., Kamruzzaman, Md., Zhu, X., Rahman, Md., Choi, K., 2013. Simulating land cover changes and their impacts on land surface temperature in Dhaka, Bangladesh. *Remote Sens* 5 (11), 5969–5998. <https://doi.org/10.3390/rs5115969>.
- Ahmed, S.S., Akter, T., Ma, Y., 2018. Green supply chain management (GSCM) performance implemented by the textile industry of Gazipur District, Dhaka. *Logistics* 2 (4), 21. <https://doi.org/10.3390/logistics2040021>.
- Arshad, S., Ahmad, S.R., Abbas, S., Asharf, A., Siddiqui, N.A., Islam, Z.U., 2022. Quantifying the contribution of diminishing green spaces and urban sprawl to urban heat island effect in a rapidly urbanizing metropolitan city of Pakistan. *Land Use Policy* 113, 105874. <https://doi.org/10.1016/j.landusepol.2021.105874>.
- Ashwini, K., Sil, B.S., 2022. Impacts of land use and land cover changes on land surface temperature over Cachar region, Northeast India—A case study. *Sustainability* 14 (21), 14087. <https://doi.org/10.3390/su142114087>.
- Bahi, H., Rhinane, H., Bensalmia, A., Fehrenbach, U., Scherer, D., 2016. Effects of urbanization and seasonal cycle on the surface urban Heat Island patterns in the coastal growing cities: A case study of Casablanca, Morocco. *Remote Sens* 8 (10), 829. <https://doi.org/10.3390/rs8100829>.
- BBS, 2011. District Statistics 2011, Dhaka. Bangladesh Bureau of Statistics, Statistics and Informatics Division (SID), Ministry of Planning, Government of The People's Republic of Bangladesh, Parishankhan Bhaban E-27/A, Agargaon, Dhaka-1207. www.bbs.gov.bd.
- BBS, 2022. District Statistics 2022, Dhaka. Bangladesh Bureau of Statistics, Statistics and Informatics Division (SID), Ministry of Planning, Government of The People's Republic of Bangladesh, Parishankhan Bhaban E-27/A, Agargaon, Dhaka-1207. www.bbs.gov.bd.
- Bhat, P.A., Shafiq, M.U., Mir, A.A., Ahmed, P., 2017. Urban sprawl and its impact on landuse/land cover dynamics of Dehradun City, India. *Int. J. Sustain. Built Environ.* 6 (2), 513–521. <https://doi.org/10.1016/j.ijbs.2017.10.003>.
- BMD, 2023. Monthly Climate Summary 2022–2023. Bangladesh Meteorological Department, Agargaon, Dhaka. <https://bmd.gov.bd/>.
- Bokaie, M., Zarkesh, M.K., Arasteh, P.D., Hosseini, A., 2016. Assessment of urban Heat Island based on the relationship between land surface temperature and land use/land cover in Tehran. *Sustain. Cities Soc.* 23, 94–104. <https://doi.org/10.1016/j.scs.2016.03.009>.
- Broge, N.H., Mortensen, J.V., 2002. Deriving green crop area index and canopy chlorophyll density of winter wheat from spectral reflectance data. *Remote Sens. Environ.* 81 (1), 45–57. [https://doi.org/10.1016/S0034-4257\(01\)00332-7](https://doi.org/10.1016/S0034-4257(01)00332-7).
- Cerqueira, M.A., Barbosa, R.V.R., Lôbo, T.P.D.S., 2024. Evolution of land surface temperature, land use land cover and their relationship: A case study of Maceió, Brazil. *Urban Clim.* 58, 102199. <https://doi.org/10.1016/j.uclim.2024.102199>.
- Chen, L., Wang, X., Cai, X., Yang, C., Lu, X., 2021. Seasonal variations of daytime land surface temperature and their underlying drivers over Wuhan, China. *Remote Sens* 13 (2), 323. <https://doi.org/10.3390/rs13020323>.
- Choudhury, D., Das, K., Das, A., 2019. Assessment of land use land cover changes and its impact on variations of land surface temperature in Asansol-Durgapur development region. *Egypt. J. Remote Sens. Space Sci.* 22 (2), 203–218. <https://doi.org/10.1016/j.ejrs.2018.05.004>.
- Choudhury, D., Das, A., Das, M., 2021. Investigating thermal behavior pattern (TBP) of local climatic zones (LCZs): A study on industrial cities of Asansol-Durgapur development area (ADDA), eastern India. *Urban Clim.* 35, 100727. <https://doi.org/10.1016/j.uclim.2020.100727>.
- Chughtai, A.H., Abbasi, H., Karas, I.R., 2021. A review on change detection method and accuracy assessment for land use land cover. *Remote Sens. Appl. Soc. Environ.* 22, 100482. <https://doi.org/10.1016/j.rsase.2021.100482>.
- Congalton, R.G., 2001. Accuracy assessment and validation of remotely sensed and other spatial information. *Int. J. Wildland Fire* 10 (4), 321. <https://doi.org/10.1071/WF01031>.
- Cui, L., Shi, J., 2012. Urbanization and its environmental effects in Shanghai, China. *Urban Clim.* 2, 1–15. <https://doi.org/10.1016/j.uclim.2012.10.008>.
- Das, N., Mondal, P., Sutradhar, S., Ghosh, R., 2021. Assessment of variation of land use/land cover and its impact on land surface temperature of Asansol subdivision. *Egypt. J. Remote Sens. Space Sci.* 24 (1), 131–149. <https://doi.org/10.1016/j.ejrs.2020.05.001>.
- De Sherbinin, A., Schiller, A., Pulsipher, A., 2007. The vulnerability of global cities to climate hazards. *Environ. Urban.* 19 (1), 39–64. <https://doi.org/10.1177/0956247807076725>.
- Dou, P., Chen, Y., 2017. Dynamic monitoring of land-use/land-cover change and urban expansion in Shenzhen using Landsat imagery from 1988 to 2015. *Int. J. Remote Sens.* 38 (19), 5388–5407. <https://doi.org/10.1080/01431161.2017.1339926>.

- Dutta, D., Rahman, A., Paul, S.K., Kundu, A., 2021. Impervious surface growth and its inter-relationship with vegetation cover and land surface temperature in peri-urban areas of Delhi. *Urban Clim.* 37, 100799. <https://doi.org/10.1016/j.uclim.2021.100799>.
- Edan, M.H., Maarouf, R.M., Hasson, J., 2021. Predicting the impacts of land use/land cover change on land surface temperature using remote sensing approach in Al Kut, Iraq. *Phys. Chem. Earth Parts A/B/C* 123, 103012. <https://doi.org/10.1016/j.pce.2021.103012>.
- Fahmy, A.H., Amin Abdelfatah, M., El-Fiky, G., 2023. Investigating land use land cover changes and their effects on land surface temperature and urban heat islands in Sharqiyah Governorate, Egypt. *Egypt. J. Remote Sens. Space Sci.* 26 (2), 293–306. <https://doi.org/10.1016/j.ejrs.2023.04.001>.
- Faisal, A.A., Kafy, Abdulla-Al, Foyezur Rahman, A.N.M., Rakib, A.A., Akter, K.S., Raikwar, V., Jahir, D.A., Ferdousi, J., Kona, M.A., 2021. Assessment and prediction of seasonal land surface temperature change using multi-temporal Landsat images and their impacts on agricultural yields in Rajshahi, Bangladesh. *Environ. Challen.* 4, 100147. <https://doi.org/10.1016/j.envc.2021.100147>.
- Faisal, A.A., Rahman, M.S., Kafy, A.-A., Roy, S., Khan, A.R., Rahman, M.M., 2022. Impact on land surface temperature and urban Heat Island, due to land use/land cover change in Dhaka metropolitan area, using remote sensing and GIS techniques. In: Singh, In R. (Ed.), *Re-Envisioning Advances in Remote Sensing*, 1st ed. CRC Press, pp. 69–102. <https://doi.org/10.1201/9781003224624-6>.
- Faruq, M.A.A., Zaman, S., Katoh, M., 2016. Analysis of forest cover changes using landsat satellite imagery: a case study of the Madhupur Sal Forest in Bangladesh. *J. Forest Plann.* 21 (2), 29–38. <https://doi.org/10.20659/jfp.21.2.29>.
- Fattah, Md.A., Morshed, S.R., 2022. Assessment of the responses of spatiotemporal vegetation changes to climatic variability in Bangladesh. *Theor. Appl. Climatol.* 148 (1–2), 285–301. <https://doi.org/10.1007/s00704-022-03943-7>.
- Gazi, Md.Y., Rahman, Md.Z., Uddin, Md.M., Rahman, F.M.A., 2021. Spatio-temporal dynamic land cover changes and their impacts on the urban thermal environment in the Chittagong metropolitan area, Bangladesh. *GeoJournal* 86 (5), 2119–2134. <https://doi.org/10.1007/s10708-020-10178-4>.
- Goldblatt, R., Addas, A., Crull, D., Maghrabi, A., Levin, G.G., Rubinyi, S., 2021. Remotely sensed derived land surface temperature (LST) as a proxy for air temperature and thermal comfort at a small geographical scale. *Land* 10 (4), 410. <https://doi.org/10.3390/land10040410>.
- Grimm, N.B., Faeth, S.H., Golubiewski, N.E., Redman, C.L., Wu, J., Bai, X., Briggs, J.M., 2008. Global change and the ecology of cities. *Science* 319 (5864), 756–760. <https://doi.org/10.1126/science.1150195>.
- Guha, S., Govil, H., 2022. Annual assessment on the relationship between land surface temperature and six remote sensing indices using landsat data from 1988 to 2019. *Geocarto Int.* 37 (15), 4292–4311. <https://doi.org/10.1080/10106049.2021.1886339>.
- Guha, S., Govil, H., Dey, A., Gill, N., 2020. A case study on the relationship between land surface temperature and land surface indices in Raipur City, India. *Geografisk Tidsskrift-Danish J. Geogr.* 120 (1), 35–50. <https://doi.org/10.1080/00167223.2020.1752272>.
- Haldar, S., Mandal, S., Bhattacharya, S., Paul, S., 2023. Dynamicity of land use/land cover (LULC) an analysis from peri-urban and rural neighbourhoods of Durgapur municipal corporation (DMC) in India. *Region. Sustain.* 4 (2), 150–172. <https://doi.org/10.1016/j.regsus.2023.05.001>.
- Halder, B., Bandyopadhyay, J., Banik, P., 2021. Monitoring the effect of urban development on urban heat island based on remote sensing and geo-spatial approach in Kolkata and adjacent areas, India. *Sustain. Cities Soc.* 74, 103186. <https://doi.org/10.1016/j.scs.2021.103186>.
- Halder, B., Bandyopadhyay, J., Al-Hilali, A.A., Ahmed, A.M., Falah, M.W., Abed, S.A., Falih, K.T., Khedher, K.M., Scholz, M., Yaseen, Z.M., 2022. Assessment of urban green space dynamics influencing the surface urban heat stress using advanced geospatial techniques. *Agronomy* 12 (9), 2129. <https://doi.org/10.3390/agronomy12092129>.
- Hasnat, G.N.T., 2022. Assessment of spatiotemporal distribution pattern of land surface temperature with incessant urban sprawl over Khulna and Rajshahi City corporations. *Environ. Challen.* 9, 100644. <https://doi.org/10.1016/j.envc.2022.100644>.
- Hass, A., Ellis, K., Reyes Mason, L., Hathaway, J., Howe, D., 2016. Heat and humidity in the City: neighborhood heat index variability in a mid-Sized City in the southeastern United States. *Int. J. Environ. Res. Public Health* 13 (1), 117. <https://doi.org/10.3390/ijerph13010117>.
- Herrera, M.D., Arreola Esquivel, M.M., Toxqui-Quilt, C., Padilla-Vivanco, A., 2019. Normalized difference indices in Landsat 5 TM satellite data. In: Johnson, R.B., Mahajan, V.N., Thibault, S. (Eds.), *Current Developments in Lens Design and Optical Engineering XX*. SPIE, p. 35. <https://doi.org/10.1117/12.2532322>.
- How Jin Aik, D., Ismail, M.H., Muharam, F.M., 2020. Land use/land cover changes and the relationship with land surface temperature using Landsat and MODIS imageries in Cameron Highlands, Malaysia. *Land* 9 (10), 372. <https://doi.org/10.3390/land9100372>.
- Hu, C., Zhang, M., Huang, G., Li, Z., Sun, Y., Zhao, J., 2024. Tracking the impact of the land cover change on the spatial-temporal distribution of the thermal comfort: insights from the Qinhuai River Basin, China. *Sustain. Cities Soc.* 116, 105916. <https://doi.org/10.1016/j.scs.2024.105916>.
- Hussain, A., Madan, R., Kamboj, V., 2022. Applications of remote sensing and GIS in hydrological and hydrogeological studies: Integrated watershed management. In: Hussain, A., Tiwari, K.C., Gupta, A. (Eds.), *Advances in Geospatial Technologies*. IGI Global, pp. 237–251. <https://doi.org/10.4018/978-1-7998-8331-9.ch012>.
- Imran, H.M., Hossain, A., Islam, A.K.M.S., Rahman, A., Bhuiyan, M.A.E., Paul, S., Alam, A., 2021. Impact of land cover changes on land surface temperature and human thermal comfort in Dhaka City of Bangladesh. *Earth Syst. Environ.* 5 (3), 667–693. <https://doi.org/10.1007/s41748-021-00243-4>.
- Jamali, A.A., Ghorbani Kalkhajeh, R., Randhir, T.O., He, S., 2022. Modeling relationship between land surface temperature anomaly and environmental factors using GEE and Giovanni. *J. Environ. Manag.* 302, 113970. <https://doi.org/10.1016/j.jenvman.2021.113970>.
- Jandaghian, Z., Colombo, A., 2024. The role of water bodies in climate regulation: insights from recent studies on urban Heat Island mitigation. *Buildings* 14 (9), 2945. <https://doi.org/10.3390/buildings14092945>.
- Jesdale, B.M., Morello-Frosch, R., Cushing, L., 2013. The racial/ethnic distribution of heat risk-related land cover in relation to residential segregation. *Environ. Health Perspect.* 121 (7), 811–817. <https://doi.org/10.1289/ehp.1205919>.
- Kafy, A.A., Rahman, Md.S., Faisal, A.A., Hasan, M.M., Islam, M., 2020. Modelling future land use land cover changes and their impacts on land surface temperatures in Rajshahi, Bangladesh. *Remote Sens. Appl. Soc. Environ.* 18, 100314. <https://doi.org/10.1016/j.rsase.2020.100314>.
- Kafy, A.A., Faisal, A.A., Rakib, A.A., Roy, S., Ferdousi, J., Raikwar, V., Kona, M.A., Fatin, S.M.A.A., 2021a. Predicting changes in land use/land cover and seasonal land surface temperature using multi-temporal landsat images in the northwest region of Bangladesh. *Heliyon* 7 (7), e07623. <https://doi.org/10.1016/j.heliyon.2021.e07623>.
- Kafy, A.A., Rakib, A.A., Akter, K.S., Rahaman, Z.A., Faisal, A.-A., Mallik, S., Nasher, N.M.R., Hossain, M.I., Ali, M.Y., 2021b. Monitoring the effects of vegetation cover losses on land surface temperature dynamics using geospatial approach in Rajshahi City, Bangladesh. *Environ. Challen.* 4, 100187. <https://doi.org/10.1016/j.envc.2021.100187>.
- Karakuş, C.B., 2019. The impact of land use/land cover (LULC) changes on land surface temperature in Sivas City Center and its surroundings and assessment of urban Heat Island. *Asia-Pac. J. Atmos. Sci.* 55 (4), 669–684. <https://doi.org/10.1007/s13143-019-00109-w>.
- Kayet, N., Pathak, K., Chakrabarty, A., Sahoo, S., 2016. Spatial impact of land use/land cover change on surface temperature distribution in Saranda Forest, Jharkhand. *Model. Earth Syst. Environ.* 2 (3), 127. <https://doi.org/10.1007/s40808-016-0159-x>.
- Li, Z.-L., Tang, B.-H., Wu, H., Ren, H., Yan, G., Wan, Z., Trigo, I.F., Sobrino, J.A., 2013. Satellite-derived land surface temperature: current status and perspectives. *Remote Sens. Environ.* 131, 14–37. <https://doi.org/10.1016/j.rse.2012.12.008>.
- Liang, X., Ji, X., Guo, N., Meng, L., 2021. Assessment of urban heat islands for land use based on urban planning: A case study in the main urban area of Xuzhou City, China. *Environ. Earth Sci.* 80 (8), 308. <https://doi.org/10.1007/s12665-021-09588-5>.
- Lillesand, T.M., Kiefer, R.W., Chipman, J.W., 2015. *Remote Sensing and Image Interpretation*, Seventh edition. Wiley.
- Liou, Y.-A., Tran, D.-P., Nguyen, K.-A., 2024. Spatio-temporal patterns and driving forces of surface urban heat island in Taiwan. *Urban Clim.* 53, 101806. <https://doi.org/10.1016/j.uclim.2024.101806>.
- Mahata, B., Sankar Sahu, S., Sardar, A., Laxmikanta, R., Maity, M., 2024. Spatiotemporal dynamics of land use/land cover (LULC) changes and its impact on land surface temperature: A case study in New Town Kolkata, eastern India. *Region. Sustain.* 5 (2), 100138. <https://doi.org/10.1016/j.regsus.2024.100138>.
- Maimaitiyiming, M., Ghulam, A., Tiyyip, T., Pla, F., Latorre-Carmona, P., Halik, Ü., Sawut, M., Caetano, M., 2014. Effects of green space spatial pattern on land surface temperature: implications for sustainable urban planning and climate change adaptation. *ISPRS J. Photogramm. Remote Sens.* 89, 59–66. <https://doi.org/10.1016/j.isprsjprs.2013.12.010>.
- Mallick, J., Kant, Y., Bharath, B.D., 2008. Estimation of land surface temperature over Delhi using Landsat-7 ETM+. *Environ. Sci. Geogr.* 12 (3), 131–140.

- Maxwell, A.E., Warner, T.A., Guillén, L.A., 2021. Accuracy assessment in convolutional neural network-based deep learning remote sensing studies—part 1: literature review. *Remote Sens* 13 (13), 2450. <https://doi.org/10.3390/rs13132450>.
- Mia, M.B., Hasan, M.A., Gazi, M.Y., Chowdhury, M., Montakim, N.A., 2022. Impact of urbanization on Landuse-Landcover, land surface temperature and urban Heat Islands using multispectral satellite images: an implication in the district towns of northwestern Bangladesh. *Dhaka Univ. J. Earth Environ. Sci.* 17–28. <https://doi.org/10.3329/dujecs.v10i3.59068>.
- Miah, M.T., Fariha, J.N., Jodder, P.K., Al Kafy, A., Raiyan, R., Usha, S.A., Hossan, J., Rahaman, K.R., 2024. Urban Heat Island and environmental degradation analysis utilizing a remote sensing technique in rapidly urbanizing South Asian cities. *World* 5 (4), 1023–1053. <https://doi.org/10.3390/world5040052>.
- Milesi, C., Elvidge, C.D., Nemani, R.R., Running, S.W., 2003. Assessing the impact of urban land development on net primary productivity in the southeastern United States. *Remote Sens. Environ.* 86 (3), 401–410. [https://doi.org/10.1016/S0034-4257\(03\)00081-6](https://doi.org/10.1016/S0034-4257(03)00081-6).
- Naim, Md.N.H., Kafy, A.-A., 2021. Assessment of urban thermal field variance index and defining the relationship between land cover and surface temperature in Chattogram city: A remote sensing and statistical approach. *Environ. Challen.* 4, 100107. <https://doi.org/10.1016/j.envc.2021.100107>.
- NourEldeen, N., Mao, K., Yuan, Z., Shen, X., Xu, T., Qin, Z., 2020. Analysis of the spatiotemporal change in land surface temperature for a long-term sequence in Africa (2003–2017). *Remote Sens* 12 (3), 488. <https://doi.org/10.3390/rs12030488>.
- Obilor, E.I., Amadi, E.C., 2018. Test for significance of Pearson's correlation coefficient. *Int. J. Innov. Math. Stat. Energy Pol.* 6 (1), 11–23.
- Patel, S., Indraganti, M., Jawarneh, R.N., 2024. Land surface temperature responses to land use dynamics in urban areas of Doha, Qatar. *Sustain. Cities Soc.* 104, 105273. <https://doi.org/10.1016/j.scs.2024.105273>.
- Rakib, A.A., Akter, K.S., Rahman, N., Arpi, S., 2020. Analyzing the pattern of land use land cover change and its impact on land surface temperature: A remote sensing approach in Mymensingh, Bangladesh. In: 1st International Student Research Conference –2020. Dhaka University Research Society (DURS), University of Dhaka, Bangladesh.
- Ramaiah, M., Avtar, R., Rahman, Md.M., 2020. Land cover influences on LST in two proposed smart cities of India: comparative analysis using spectral indices. *Land* 9 (9), 292. <https://doi.org/10.3390/land9090292>.
- Rasul, A., Balzter, H., Smith, C., 2015. Spatial variation of the daytime Surface Urban Cool Island during the dry season in Erbil, Iraqi Kurdistan, from Landsat 8. *Urban Clim.* 14, 176–186. <https://doi.org/10.1016/j.uclim.2015.09.001>.
- Rousta, I., Sarif, M.O., Gupta, R.D., Olafsson, H., Ranagalage, M., Murayama, Y., Zhang, H., Mushore, T.D., 2018. Spatiotemporal analysis of land use/land cover and its effects on surface urban Heat Island using Landsat data: a case study of Metropolitan City Tehran (1988–2018). *Sustainability* 10 (12), 4433. <https://doi.org/10.3390/su10124433>.
- Roy, B., Bari, E., Nipa, N.J., Ani, S.A., 2021. Comparison of temporal changes in urban settlements and land surface temperature in Rangpur and Gazipur Sadar, Bangladesh after the establishment of city corporation. *Remote Sens. Appl. Soc. Environ.* 23, 100587. <https://doi.org/10.1016/j.rsase.2021.100587>.
- Saha, S., Saha, A., Das, M., Saha, A., Sarkar, R., Das, A., 2021. Analyzing spatial relationship between land use/land cover (LULC) and land surface temperature (LST) of three urban agglomerations (UAs) of eastern India. *Remote Sens. Appl. Soc. Environ.* 22, 100507. <https://doi.org/10.1016/j.rsase.2021.100507>.
- Saha, J., Ria, S.S., Sultana, J., Shima, U.A., Hasan Seyam, M.M., Rahman, M.M., 2024. Assessing seasonal dynamics of land surface temperature (LST) and land use land cover (LULC) in Bhairab, Kishoreganj, Bangladesh: A geospatial analysis from 2008 to 2023. *Case Stud. Chem. Environ. Eng.* 9, 100560. <https://doi.org/10.1016/j.csee.2023.100560>.
- Santamouris, M., 2015. Analyzing the heat island magnitude and characteristics in one hundred Asian and Australian cities and regions. *Sci. Total Environ.* 512–513, 582–598. <https://doi.org/10.1016/j.scitotenv.2015.01.060>.
- Santhosh, L.G., Shilpa, D.N., 2023. Assessment of LULC change dynamics and its relationship with LST and spectral indices in a rural area of Bengaluru district, Karnataka India. *Remote Sens. Appl. Soc. Environ.* 29, 100886. <https://doi.org/10.1016/j.rsase.2022.100886>.
- Shahfahad, Talukdar, S., Naikoo, M.W., Rihan, M., Mohammad, P., Rahman, A., 2024. Seasonal dynamics of land surface temperature and urban thermal comfort with land use land cover pattern in semi-arid Indian cities: insights for sustainable Urban Management. *Urban Clim.* 57, 102105. <https://doi.org/10.1016/j.uclim.2024.102105>.
- Silva López, J.O., Salas López, R., Rojas Briceño, N.B., Gómez Fernández, D., Terrones Murga, R.E., Iliquin Trigos, D., Barboza Castillo, E., Oliva Cruz, M., Barrena Gorbillon, M.A., 2022. Analytic hierarchy process (AHP) for a landfill site selection in Chachapoyas and Huancas (NW Peru): modeling in a GIS-RS environment. *Adv. Civil Eng.* 2022 (1), 9733322. <https://doi.org/10.1155/2022/9733322>.
- Singh, P., Kikon, N., Verma, P., 2017. Impact of land use change and urbanization on urban heat island in Lucknow city, Central India. A remote sensing based estimate. *Sustain. Cities Soc.* 32, 100–114. <https://doi.org/10.1016/j.scs.2017.02.018>.
- Sresto, M.A., Siddika, S., Fattah, Md.A., Morshed, S.R., Morshed, Md.M., 2022. A GIS and remote sensing approach for measuring summer-winter variation of land use and land cover indices and surface temperature in Dhaka district, Bangladesh. *Heliyon* 8 (8), e10309. <https://doi.org/10.1016/j.heliyon.2022.e10309>.
- Theodorou, P., 2022. The effects of urbanisation on ecological interactions. *Curr. Opin. Insect Sci.* 52, 100922. <https://doi.org/10.1016/j.cois.2022.100922>.
- Tomlinson, C.J., Chapman, L., Thornes, J.E., Baker, C.J., 2011. Including the urban heat island in spatial heat health risk assessment strategies: A case study for Birmingham, UK. *Int. J. Health Geogr.* 10 (1), 42. <https://doi.org/10.1186/1476-072X-10-42>.
- Tucker, C.J., 1979. Red and photographic infrared linear combinations for monitoring vegetation. *Remote Sens. Environ.* 8 (2), 127–150.
- Verma, R., Garg, P.K., 2021. Mapping the spatiotemporal changes of land use/land cover on the urban Heat Island effect by open source data: A case study of Lucknow, India. *J. Indian Soc. Remote Sens.* 49 (11), 2655–2671. <https://doi.org/10.1007/s12524-021-01421-7>.
- Vohra, R., Kumar, A., Jain, R., Hemanth, D.J., 2024. Analysis and prediction of land surface temperature with increasing urbanisation using satellite imagery. *Heliyon* 10 (22), e40378. <https://doi.org/10.1016/j.heliyon.2024.e40378>.
- Wang, A., Dai, Y., Zhang, M., Chen, E., 2025. Exploring the cooling intensity of green cover on urban heat island: A case study of nine main urban districts in Chongqing. *Sustain. Cities Soc.* 124, 106299. <https://doi.org/10.1016/j.scs.2025.106299>.
- Weng, Q., Lu, D., Schubring, J., 2004. Estimation of land surface temperature–vegetation abundance relationship for urban heat island studies. *Remote Sens. Environ.* 89 (4), 467–483. <https://doi.org/10.1016/j.rse.2003.11.005>.
- Yadav, N., Sharma, C., Peshin, S.K., Masiwal, R., 2017. Study of intra-city urban heat island intensity and its influence on atmospheric chemistry and energy consumption in Delhi. *Sustain. Cities Soc.* 32, 202–211. <https://doi.org/10.1016/j.scs.2017.04.003>.
- Zhang, Y., Tao, Y., Gu, X., Zhang, Y., Yu, S., Zhang, W., Li, X., 2006. Land surface temperature retrieval from CBERS-02 IRMSS thermal infrared data and its applications in quantitative analysis of urban Heat Island effect. *Natl. Remote Sens. Bull.* 10 (5), 789–797. <https://doi.org/10.11834/jrs.200605117>.
- Zhang, Y., Odeh, I.O.A., Ramadan, E., 2013. Assessment of land surface temperature in relation to landscape metrics and fractional vegetation cover in an urban/peri-urban region using Landsat data. *Int. J. Remote Sens.* 34 (1), 168–189. <https://doi.org/10.1080/01431161.2012.712227>.
- Zhang, M., Yigit, I., Adigüzel, F., Hu, C., Chen, E., Siyavuş, A.E., Elmastaş, N., Ustuner, M., Kaya, A.Y., 2024. Impact of urban surfaces on microclimatic conditions and thermal comfort in Burdur, Türkiye. *Atmosphere* 15 (11), 1375. <https://doi.org/10.3390/atmos15111375>.
- Zhao, J., Hu, C., Li, Z., Zhang, M., Fan, H., Li, K., Yuan, R., 2025. Investigating the quantitative impact of the vegetation indices on the urban thermal comfort based on machine learning: A case study of the Qinhuai River Basin, China. *Sustain. Cities Soc.* 125, 106357. <https://doi.org/10.1016/j.scs.2025.106357>.
- Zhou, D., Zhao, S., Liu, S., Zhang, L., Zhu, C., 2014. Surface urban heat island in China's 32 major cities: spatial patterns and drivers. *Remote Sens. Environ.* 152, 51–61. <https://doi.org/10.1016/j.rse.2014.05.017>.
- Zhou, D., Xiao, J., Bonafoni, S., Berger, C., Deilami, K., Zhou, Y., Frolking, S., Yao, R., Qiao, Z., Sobrino, J.A., 2018. Satellite remote sensing of surface urban Heat Islands: progress, challenges, and perspectives. *Remote Sens* 11 (1), 48. <https://doi.org/10.3390/rs11010048>.
- Zhu, H., Wang, A., Wang, P., Hu, C., Zhang, M., 2025. Spatiotemporal dynamics and response of land surface temperature and kernel normalized difference vegetation index in Yangtze River Economic Belt, China: multi-method analysis. *Land* 14 (3), 598. <https://doi.org/10.3390/land14030598>.